Nearly Optimal Bounds for Orthogonal Trace-Sum Maximization

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Abstract

Orthogonal trace-sum maximization (OTSM) problems arise in a wide range of data processing applications, including canonical correlation analysis and cryogenic electron microscopy. Despite their practical importance, the existing theoretical understanding of these problems remains incomplete. In this paper, we show that the generalized power method (GPM) converges *linearly* to the global optimal solution with high probability under an additive sub-Gaussian noise model, provided that the noise level satisfies the nearly optimal bound $O(\sqrt{n/\log n})$. In addition, we prove that the semidefinite programming (SDP) relaxation of OTSM is tight and admits a unique optimal solution under the same noise regime, improving upon the best known theoretical guarantee of $O(n^{1/4})$. We also establish an information theoretic minimax lower bound for rectangular Stiefel recovery, showing that our sufficient noise scale is optimal up to a $\sqrt{\log n}$ factor in the nondegenerate fixed dimensional setting. Extensive numerical experiments further demonstrate that our theoretical predictions closely match empirical observations.

1 Introduction

Orthogonal trace–sum maximization (OTSM) problems constitute an important class of data processing problems with widespread applications across science and engineering, including canonical correlation analysis (CCA) [14, 16], Procrustes analysis [12, 17, 13], and cryogenic electron microscopy (cryo-EM) [26, 24], among many others. A prototype of OTSM problems takes the form

$$\max \quad \sum_{(i,j)\in\Omega} \operatorname{tr}(O_i^\top C_{ij} O_j) \qquad \text{s.t.} \quad O_i \in O_{d_i,r},\ i \in [n] \tag{OTSM}$$

where $\Omega \subseteq [n] \times [n]$ denotes the observation set, $O_{d_i,r} \coloneqq \{O \in \mathbb{R}^{d_i \times r} \mid O^\top O = I_r\}$ (with $d_i \geq r$) denotes the Stiefel manifold, and the matrices $C_{ij} = C_{ji}^\top \in \mathbb{R}^{d_i \times d_j}$ encode pairwise information.

In this paper, we impose a generative model on the observed matrices $\{C_{ij}\}_{(i,j)\in\Omega}$. A commonly adopted assumption in the literature posits the existence of a collection of ground-truth signals $\{G_i^\star\}_{i=1}^n$ together with noise matrices $\{W_{ij}\}_{(i,j)\in\Omega}$, such that

$$C_{ij} = G_i^\star G_j^{\star\top} + \sigma W_{ij}, \qquad (i,j) \in \Omega, \tag{1}$$

where $\sigma \geq 0$ controls the noise level. Throughout, $W = W^\top$ has mutually independent, mean-zero scalar entries on and above its main diagonal, with sub-Gaussian norms normalized so that $\|W_{ab}\|_{\psi_2} \leq 1$; entries below the diagonal are determined by symmetry. Any fixed noise scale can be absorbed into σ. In the sampled model, the Bernoulli observation masks are independent of W. The OTSM problem with assumption (1) is commonly referred to as the Stiefel manifold synchronization problem. Under this model, the goal is to recover the ground-truth signals $\{G_i^\star\}_{i=1}^n$ from noisy pairwise observations. For notational convenience, we stack $\{C_{ij}\}$ and $\{W_{ij}\}$ into block matrices C and W, whose (i,j)-th blocks are C_{ij} and W_{ij} if $(i,j) \in \Omega$ and 0 otherwise.

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We study semidefinite relaxation (SDR) and the generalized power method (GPM). The SDR of (OTSM) is [29]

$$\max_{U \succeq 0} \quad \langle C, U \rangle \qquad \text{s.t.} \quad U_{ii} \preceq I_{d_i},\ \mathrm{tr}(U_{ii}) = r,\ \forall i \in [n], \tag{2}$$

which lifts the problem to a positive semidefinite matrix U and drops the constraint $\mathrm{rank}(U) \leq r$.

In contrast, the generalized power method (GPM) directly operates on the original nonconvex feasible set and scales favorably to large problem instances. In its simplest form, GPM iteration is given by

$$G^{t+1} = \pi^n(CG^t) \quad \text{for } t = 0, 1, \ldots,$$

where $G^t = (G_1^t; \ldots; G_n^t)$, and $\pi^n : \mathbb{R}^{(\sum_i d_i) \times r} \to O_{d_1,r} \times \cdots \times O_{d_n,r}$ denotes the blockwise projection onto the product of Stiefel manifolds. Each projection step amounts to computing the polar factor from a thin singular value decomposition of the corresponding block, making GPM computationally efficient and easy to implement.

A central question in the study of problem (OTSM) concerns

> when the SDR is tight, and under what conditions the GPM converges to a global optimum, as well as at what rate.

Intuitively, one expects that when the observation graph Ω is sufficiently informative and the noise level σ is not too large, near-exact recovery of the ground-truth solution is possible via either SDR or GPM. Characterizing the precise regimes—both in terms of noise magnitude and sampling density—under which such recovery guarantees hold is a central focus of this line of research.

Despite the widespread use of the generative model (OTSM), the theoretical understanding of OTSM remains incomplete. For the SDR (2), the best existing tightness guarantee, established by [29], requires $\sigma = O(n^{1/4})$ in the fully observed case $p = 1$, which falls well short of the empirically observed $O(\sqrt{n})$ noise regime. Beyond the generative setting, [3] proves a general approximation guarantee for SDR; however, its approximation factor $1/(2\sqrt{2})$ is far from tightness and does not account for exact recovery phenomena. The situation is even more limited for the generalized power method: existing results [30] guarantee convergence only to a stationary point, without establishing convergence rates or statistical recovery guarantees for the ground-truth signal. Existing minimax lower bounds concern phase synchronization and square orthogonal/rotation synchronization [10, 11]; to our knowledge, a corresponding bound for general heterogeneous rectangular Stiefel factors under (1) has not been available.

As a result, a substantial gap persists between theory and practice. Empirical evidence suggests that both SDR and GPM remain effective at significantly higher noise levels, with successful recovery observed when $\sigma = O(\sqrt{n})$ (see Section 4). This discrepancy highlights the need for sharper theoretical analyses capable of explaining the strong empirical performance of these methods.

1.1 Main Contributions

In this paper, we nearly close the gap between existing theoretical guarantees and empirical observations by establishing linear convergence of the GPM and tightness of the SDR under substantially weaker noise conditions $\sigma = O(\sqrt{n/\log n})$. Our analysis is built upon a careful application of the leave-one-out technique [31, 1, 20], which allows us to control delicate dependencies arising from nonconvexity and randomness. Our main contributions can be summarized as follows.

- **C.1 Linear convergence of GPM.** We show that, in the fully observed case $p = 1$, the GPM with spectral initialization converges linearly to the global optimal solution of (OTSM) with high probability, provided that $\sigma = O(\sqrt{n/\log n})$.
- **C.2 Nearly optimal bounds.** We significantly improve existing guarantees for the SDR by proving tightness in the nearly optimal noise regime $\sigma = O(\sqrt{n/\log n})$, improving upon the previous best bound $\sigma = O(n^{1/4})$.
- **C.3 Random observation model.** We further extend all of the above results to the random observation setting, where each pairwise measurement is observed independently with prob-

ability $p \gtrsim \log n/\sqrt{n}$. In this case, we show that both linear convergence of GPM and tightness of SDR hold when $\sigma = O(\sqrt{np/\log n})$.

C.4 Information-theoretic lower bound. We prove a minimax lower bound for rectangular Stiefel recovery, accounting for column-space and rotational degrees of freedom. In the nondegenerate fixed-dimensional setting, our sufficient noise scale is optimal for statistical recovery up to $\sqrt{\log n}$.

1.2 Related Work

Due to space limitations, we defer the discussion of related work to Section A.

Notation. For notational simplicity, we define $d := \frac{1}{n}\sum_{i=1}^{n} d_i$, which is assumed to be independent of n. Define $O_{d,r}^n = O_{d_1,r} \times \cdots \times O_{d_n,r}$. We use $O(d) = O_{d,d}$ to denote the set of orthogonal matrices with dimension d. We represent the collection of variables $G_i \in O_{d_i,r}$ by stacking them into a single matrix

$$G = (G_1; \cdots; G_n) \in \mathbb{R}^{nd\times r}.$$

For any matrix $X \in \mathbb{R}^{nd\times nd}$, we denote by $X_{ij} \in \mathbb{R}^{d_i\times d_j}$ its (i,j)-th block. Throughout the paper, the notation $x \gtrsim y$ and $x \lesssim y$ means that there exists a constant $c > 0$, independent of n and p, such that $x \geq c\,y$ and $x \leq c\,y$, respectively. Besides, we shall use $\mathrm{BlkDiag}(X_1, \ldots, X_K)$ to denote the block diagonal matrix whose diagonal blocks are $X_1, \ldots, X_K$.

2 Preliminaries

2.1 Motivation

CCA, Procrustes analysis, and cryo-EM motivate OTSM-type objectives, but their complete raw data-generating mechanisms can be more structured than independent additive matrix noise. Our additive sub-Gaussian model is a *theoretically tractable abstraction*. Similar abstractions are standard in related theory: [19] and [20] use independent additive Gaussian noise for orthogonal synchronization, while [29] uses an additive Gaussian model for general Stiefel OTSM. Independent additive Gaussian image noise is also standard in cryo-EM likelihood analyses [25], although pairwise quantities derived from such images need not retain entrywise independence.

A direct application satisfying our assumptions is normalized Gaussian collective matrix factorization (CMF) [15]. Its orthogonal unit-strength specialization is $C_{ij} = G_i^\star G_j^{\star\top} + \sigma W_{ij}$ with $G_i^\star \in O_{d_i,r}$, where the relation matrices may have heterogeneous dimensions and the upper-triangular noise blocks have independent Gaussian entries. This is precisely our generative model.[1]

Under Gaussian noise, maximum likelihood minimizes the sum of squared Frobenius errors between C_{ij} and $G_i G_j^\top$. Since the squared Frobenius norm of $G_i G_j^\top$ equals r, expanding the objective leaves the variable-dependent term $-2\,\mathrm{tr}(G_i^\top C_{ij} G_j)$. Therefore, the maximum-likelihood estimator is exactly OTSM.

We present normalized Gaussian CMF as a direct application, while retaining CCA, Procrustes analysis, and cryo-EM as broader structural motivations. The general CMF model with relation-specific scaling matrices D_{ij} is outside the present scope.

2.2 Algorithm

In this section, we review the spectral initialization procedure and the generalized power method (GPM) for solving problem (OTSM).

Spectral initialization. Spectral initialization is a relaxation-based technique that constructs an initial point by exploiting the spectral structure of the information matrix C. Specifically, we compute the leading r eigenvectors of C, denoted by $\widetilde{G} \in \mathbb{R}^{nd\times r}$, and then project $\widetilde{G}$ onto the product of Stiefel manifolds. Let

$$\pi^n : \mathbb{R}^{nd\times r} \to O_{d_1,r} \times \cdots \times O_{d_n,r}$$

[1] For symmetric self-relation blocks, use independent upper-triangular Gaussian noise with diagonal variance twice the off-diagonal variance. Their likelihood contributes half the squared Frobenius error, matching the symmetric OTSM objective. The recovery guarantees also require the sampling assumptions stated below.

Algorithm 1 GPM for Stiefel manifold synchronization

Input: Measurement matrix $C \in \mathbb{R}^{nd\times nd}$, size r
Initialization: Compute the top r eigenvectors $\widetilde{G} \in \mathbb{R}^{nd\times r}$ of C and set $G^0 = \pi^n(\widetilde{G})$
for $t = 0, 1, \cdots$ **do**
  Set $G^{t+1} = \pi^n(CG^t)$
  Stop if $\|G^{t+1} - G^t\|_F$ is small enough
end for

denote the blockwise projection operator. For each block i, the projection $\pi_i : \mathbb{R}^{d_i\times r} \to O_{d_i,r}$ is

$$\pi_i(\widetilde{G}_i) = P_i Q_i^\top,$$

where $P_i \in \mathbb{R}^{d_i\times r}$ and $Q_i \in \mathbb{R}^{r\times r}$ are the left and right singular vectors of $\widetilde{G}_i$, respectively.

Generalized power method. The generalized power method was introduced in [18] as an iterative procedure for maximizing a convex function over a compact feasible set. At each iteration, GPM maximizes an affine minorant of the objective function at the current iterate over the feasible set to produce the next iterate. In the context of problem (OTSM), the standard GPM update takes the form

$$G^{t+1} = \pi^n\left(\left(I + \frac{\alpha}{n}C\right)G^t\right),$$

where $\alpha \in (0, \infty]$ is a stepsize parameter. In Algorithm 1, we focus on the special case $\alpha = \infty$, which reduces to a power-iteration-type update analogous to the classical power method in the vector setting.

2.3 Problem Setting

We assume an Erdős–Rényi observation model: for every unordered block pair $i \le j$, a mask $e_{ij} \sim \text{Bernoulli}(p)$, $p \in (0, 1]$, is drawn independently; set $e_{ji} = e_{ij}$ and $\Omega = \{(i, j) : e_{ij} = 1\}$. All masks are independent of the noise. Unless stated otherwise, our algorithmic guarantees use the spectral initialization in Algorithm 1. Throughout the paper, the block dimensions d_i and rank r are treated as absolute constants, except in Theorem 3.2 and its proof.

Since the measurements $\{C_{ij} : (i, j) \in \Omega\}$ in (OTSM) are invariant under multiplication of a common rotation in $O(r)$ to the target signal $G^\star$, we can only identify $G^\star$ up to a global rotation. This motivates us to define the following quotient space.

Definition 2.1 (Quotient space)**.** For any $A, B \in \mathbb{R}^{nd\times r}$, we define an equivalence relation $\sim$ by

$$A \sim B \;\Leftrightarrow\; \exists Q \in O(r), \text{ such that } A = BQ.$$

In the quotient space $\mathbb{R}^{nd\times r}/\sim$, we define metric d_F by

$$d_F([A], [B]) = d_F(A, B) = \min_{Q\in O(r)} \|A - BQ\|_F.$$

The subspace $(O^n_{d,r}/\sim, d_F)$ is a complete metric space. Throughout the paper, uniqueness of a factor solution is understood in this quotient space, namely up to a common rotation in $O(r)$. Uniqueness of the lifted SDP matrix $GG^\top$ is ordinary matrix uniqueness.

In our analysis, we also need the following concept of block-wise norm and block-wise distance.

Definition 2.2 (Block-wise norm and distance)**.** The block-wise norm $\|\cdot\|_\infty$ on $\mathbb{R}^{nd\times r}$ is defined as

$$\|A\|_\infty = \max_{i\in[n]} \|A_i\|_F, \quad \forall A \in \mathbb{R}^{nd\times r}$$

where $A_i \in \mathbb{R}^{d_i\times r}$ is the i-th block of A. The block-wise distance d_∞ is defined as

$$d_\infty(A, B) = \min_{Q\in O(r)} \|A - BQ\|_\infty, \quad \forall A, B \in \mathbb{R}^{nd\times r}.$$

3 Main Results

We first summarize the algorithmic guarantees, followed by a statistical minimax lower bound.

Theorem 3.1 (Overall)**.** *Suppose that the block dimensions d_i and rank r are absolute constants, and that the sampling ratio p and noise level σ satisfy*

$$p \gtrsim \frac{\log n}{\sqrt{n}} \quad \text{and} \quad \sigma \lesssim \sqrt{\frac{np}{\log n}}. \tag{3}$$

Then, with probability at least $1 - O(n^{-2})$, the following statements hold:

(1) The spectrally initialized GPM converges linearly to the unique globally optimal solution G^∞, up to a common rotation, of (OTSM)*, in the sense*

$$d_F\big(G^{t+1}, G^\infty\big) \le \frac{1}{2}\, d_F\big(G^t, G^\infty\big), \qquad \forall\, t \ge 0.$$

(2) The globally optimal solution G^∞ satisfies

$$d_F(G^\infty, G^\star) \lesssim \frac{1+\sigma}{\sqrt{p}},\ d_\infty(G^\infty, G^\star) \lesssim (1+\sigma)\sqrt{\frac{\log n}{np}}.$$

(3) The semidefinite relaxation (2) *is tight, and $G^\infty (G^\infty)^\top$ is the unique optimal solution to* (2)*.*

In the fully observed case $p = 1$, it yields recovery guarantees under $\sigma = O(\sqrt{n/\log n})$, substantially improving the previous best bound $\sigma = O(n^{1/4})$ [29].

Part (1) establishes global linear convergence of spectrally initialized GPM to the global optimum G^∞, despite the nonconvexity of OTSM. Part (2) provides Frobenius and blockwise infinity-norm recovery bounds, which match the optimal rates known for phase synchronization [31] and orthogonal group synchronization [19] in the fully observed setting. Part (3) shows that the SDP relaxation is tight in the same noise regime and has a unique optimal solution. Together, these results give a unified explanation for the empirical success of both GPM and SDR under $\sqrt{np/\log n}$ noise levels.

Theorem 3.2 (Information-theoretic lower bound)**.** *Consider the Gaussian specialization of* (1) *with $n \ge 2$, $p \in (0, 1]$, and $\sigma > 0$, under the independent observation masks above. Write $h_i = r(d_i - r)$, $h_{\max} = \max_i h_i$, and $q = r(r-1)/2$. There are universal constants $c, c_0 > 0$ such that, if $\sigma^2 \max\{h_{\max}, q\} \le c_0 np$, then*

$$\inf_{\widehat{G}} \sup_{G^\star \in O^n_{d,r}} \mathbb{E}_{G^\star}\left[\frac{d_F^2(\widehat{G}, G^\star)}{n}\right] \ \ge\ c\,\frac{\sigma^2}{np}\left(\frac{1}{n}\sum_{i=1}^{n} r(d_i - r) + \frac{r(r-1)}{2}\right). \tag{4}$$

The infimum is over all estimators based on the observed masks and measurements. The bound remains valid with observed diagonal blocks; the proof is in Appendix E.

When $h_{\max}, q$ are fixed and $n^{-1}\sum_i h_i + q$ is bounded away from zero, Theorem 3.2 gives positive normalized risk at $\sigma = c_1\sqrt{np}$ for small fixed $c_1 > 0$. Our sufficient condition $\sigma \lesssim \sqrt{np/\log n}$ is therefore nearly information-theoretically optimal for recovery, up to $\sqrt{\log n}$. This logarithmic factor comes from high-probability blockwise bounds and union bounds over blocks and polynomially many leave-one-out iterates.

3.1 Full Observation Model

We first consider the fully observed setting, namely $p = 1$, where the observation set satisfies $\Omega = [n] \times [n]$. In this case, the data matrix takes the form

$$C = G^\star G^{\star\top} + \sigma W. \tag{5}$$

The results for the randomly observed setting will be presented in the next section.

3.1.1 Leave-One-Out Technique

In this section, we explain how the leave-one-out (LOO) technique is employed in our analysis. As will become clear, the main technical challenge lies in deriving a sharp bound for $\|WG^t\|_\infty$.

> *Core difficulty.* Due to the mechanisms of spectral initialization and the GPM updates, all iterates $\{G^t\}_{t\geq 0}$ are intrinsically dependent on the noise matrix W.

This dependence renders many standard concentration inequalities inapplicable, as they typically rely on independence between random variables. A naive bound of the quantity $\|WG^t\|_\infty$ obtained via

$$\|WG^t\|_\infty = \max_{i\in[n]} \|w_i^\top G^t\|_F \leq \max_{i\in[n]} \|w_i\|_2 \, \|G^t\|_F \lesssim n \tag{6}$$

is too crude for our purposes. Here, $w_i \in \mathbb{R}^{nd\times d_i}$ denotes the i-th block column of W, and the last inequality follows from the high-probability bound $\|w_i\|_2 \lesssim \sqrt{n}$ together with $\|G^t\|_F = \Theta(\sqrt{n})$.

To overcome this difficulty, we adopt the leave-one-out technique by introducing the following auxiliary objects:

- $W^{(m)} \in \mathbb{R}^{nd\times nd}$, obtained from W by setting the m-th block row and block column to 0;
- $C^{(m)} = G^\star G^{\star\top} + \sigma W^{(m)}, \;\; \Delta W^{(m)} = W - W^{(m)}$;
- $\{G^{t,m}\}_{t\geq 0}$, the sequence generated by the GPM with C replaced by $C^{(m)}$, and $\widetilde{G}^{(m)}$, the matrix formed by the leading r eigenvectors of $C^{(m)}$, normalized so that $\|\widetilde{G}^{(m)}\|^2 = nr$.

Let $d_F(G^t, G^{t,m}) = \min_{Q\in O(r)} \|G^tQ - G^{t,m}\|_F$. By the triangle inequality, we obtain

$$\begin{aligned}\|w_m^\top G^t\|_F &\leq \|w_m^\top G^{t,m}\|_F + \|w_m^\top (G^tQ - G^{t,m})\|_F \\ &\leq \|\Delta W^{(m)} G^{t,m}\|_F + \|w_m\|_2 \, d_F(G^t, G^{t,m}),\end{aligned}$$

where the second inequality uses the fact that $w_m^\top G^{t,m}$ is a block of $\Delta W^{(m)} G^{t,m}$.

Crucially, $\Delta W^{(m)}$ is independent of $G^{t,m}$, which allows us to apply concentration inequalities to show that $\|\Delta W^{(m)} G^{t,m}\|_F = O(\sqrt{n\log n})$ with high probability (see Lemma C.2). Moreover, since C and $C^{(m)}$ differ only in a single block row and column, we can show that $d_F(G^t, G^{t,m}) = O(1)$ uniformly in t (see Theorem C.10). Combining these bounds with $\|w_m\|_2 \lesssim \sqrt{n}$, we conclude that

$$\|WG^t\|_\infty \lesssim \sqrt{n\log n},$$

which is much sharper than the naive bound given by (6), and is a key estimate underpinning our convergence and recovery analysis.

Roadmap of the Proofs. We establish two main results under the noise regime $\sigma = O(\sqrt{n/\log n})$: the linear convergence of GPM and the tightness of the SDP relaxation. The proof of GPM convergence follows a contractive-mapping argument.

- We first show that spectral initialization produces an initial point $\widetilde{G}$ that is close to $G^\star$ in both the Frobenius distance d_F and the blockwise distance d_∞; see Theorem 3.3.
- We then prove that the GPM update map $\mathcal{P}_C$ is locally contractive in a neighborhood of $G^\star$; see Lemma C.3.
- Next, we show that the entire GPM trajectory remains within this contractive neighborhood; see Theorem 3.5. The main technical challenge is to obtain a sharp uniform bound on $\|WG^t\|_\infty$ along the iterates. To handle the dependence between G^t and the noise matrix W, we employ the leave-one-out technique and introduce auxiliary sequences $G^{t,m}$.
- Combining the local contractivity with the invariance of the trajectory inside the contractive region, we show that the GPM iterates form a linearly convergent sequence.
- Finally, the tightness of the SDP relaxation follows by applying the dual-certificate condition in Lemma 3.7 to the limit point G^∞.

3.1.2 Spectral Initialization

We begin with the following theorem, which guarantees that the spectral initialization produces a nearly optimal estimate.

Theorem 3.3 (Initialization)**.** *Let $\widetilde{G} \in \mathbb{R}^{nd\times r}$ be the top r leading eigenvectors of C, $\|\widetilde{G}\|_F = \sqrt{nr}$. Then there exists absolute constants $c_0, C_0 > 0$, such that when $\sigma \le c_0\sqrt{n/\log n}$, with probability at least $1 - O(n^{-2})$, the following statements hold:*

$$d_F(\widetilde{G}, G^\star) \le C_0\sigma, \quad d_\infty(\widetilde{G}, G^\star) \le C_0\sigma\sqrt{\log n/n}. \tag{7}$$

Theorem 3.3 shows that the spectral initialization lies within a small neighborhood of the ground-truth solution $G^\star$, which is a key prerequisite for establishing fast global convergence of the generalized power method. In fact, the size of the guaranteed neighborhood matches, up to constant factors, that of the global optimum characterized in Theorem 3.1, although the constants are generally conservative. This behavior is consistent with existing results for orthogonal synchronization [20], where spectral methods are shown to achieve nearly optimal statistical accuracy for synchronization over orthogonal and permutation groups.

3.1.3 Local Concentration and Convergence

A central step in our analysis is to show that the iteration mapping

$$\mathcal{P}_C : O_{d,r}^n \to O_{d,r}^n, \qquad \mathcal{P}_C(G) = \pi^n(CG/n)\,,$$

is locally contractive in a suitable neighborhood of the ground-truth solution. That is, as shown in Lemma C.3, there exist some constant $\rho \in (0,1)$ such that

$$d_F(\mathcal{P}_C(G),\, \mathcal{P}_C(G')) \le \rho\, d_F(G, G'). \tag{8}$$

Then, we can refine the accuracy of the projected spectral initialization $G^0 = \pi^n(\widetilde{G})$.

Proposition 3.4. *Under the conditions of Theorem 3.3, with probability at least $1 - O(n^{-2})$,*

$$d_F(G^0, G^\star) \lesssim \sigma, \quad d_\infty(G^0, G^\star) \lesssim \sigma\sqrt{(\log n)/n}. \tag{9}$$

The next theorem shows that the sequence generated by GPM remains within a neighborhood of the ground truth where the iteration map $\mathcal{P}_C$ is contractive. We refer to this neighborhood as the *contractive region*.

Theorem 3.5 (Contractive region)**.** *Assume $\sigma \lesssim \sqrt{n/\log n}$. Then, with probability at least $1 - O(n^{-2})$, for all $t \in \{0, 1, \ldots, 3n^2 - 1\}$,*

$$d_F(G^t, G^\star) \le k_3\sqrt{n}, \quad d_F(G^{t,m}, G^t) \le k_1,\ \forall m \in [n], \quad \|WG^t\|_\infty \le k_2\sqrt{n\log n}, \tag{10}$$

where $k_1, k_2, k_3 > 0$ are absolute constants.

The theorem controls only a finite sequence of $3n^2$ iterates, rather than directly establishing that the entire infinite sequence lies in the contractive region. This avoids a union bound over infinitely many random events. The choice $3n^2$ is technical; other sufficiently long polynomial horizons can be used with corresponding adjustments to the concentration bounds. At the end of this horizon, the iterates remain sufficiently close to $G^\star$, while contraction has made successive differences sufficiently small. The deterministic continuation argument in Theorem C.12 then keeps G^{3n^2} and every later iterate in the contractive region.

Since the iterates G^t remain in the contractive region, the mapping $\mathcal{P}_C$ is contractive along the trajectory of the algorithm. As a consequence, $\{G^t\}$ forms a Cauchy sequence in the quotient space $(O_{d,r}^n/\sim, d_F)$. This leads to the following convergence result.

Theorem 3.6. *Under the conditions of Theorem 3.5, with probability at least $1 - O(n^{-2})$,*

$$d_F(G^{t+1}, G^t) \le \frac{1}{2} d_F(G^t, G^{t-1}), \qquad \forall\, t \ge 1. \tag{11}$$

Moreover, in the metric space $(O_{d,r}^n/\sim, d_F)$, the equivalence classes $[G^t]$ converge linearly to a limit $[G^\infty]$ at rate $1/2$, namely,

$$d_F(G^{t+1}, G^\infty) \le \frac{1}{2} d_F(G^t, G^\infty), \qquad \forall\, t \ge 0, \tag{12}$$

where $G^\infty \in O_{d,r}^n$ is a fixed point of the mapping $\mathcal{P}_C$.

3.1.4 Verifying Optimality

To verify the optimality of G^∞, we invoke a dual certificate for the semidefinite relaxation. The following optimality conditions are adapted from [30].

Lemma 3.7. *Let $O = (O_1; \cdots; O_n) \in O_{d,r}^n$. Suppose there exist symmetric matrices $\Lambda_i \in \mathbb{R}^{r\times r}$, $i \in [n]$, such that $(CO)_i = O_i\Lambda_i$. Let τ_i denote the smallest eigenvalue of Λ_i, and define*

$$L(O,\Lambda) = \mathrm{BlkDiag}\Big(O_1\Lambda_1O_1^\top + \tau_1(I_{d_1} - O_1O_1^\top), \cdots, O_n\Lambda_nO_n^\top + \tau_n(I_{d_n} - O_nO_n^\top)\Big) - C.$$

If $L(O,\Lambda)$ is positive semidefinite, then O is a globally optimal solution of (OTSM)*. Moreover, if $\mathrm{rank}(L(O,\Lambda)) = nd - r$, then O is the unique global maximizer of* (OTSM)*, up to a common rotation, and $OO^\top$ is the unique global maximizer of the SDP relaxation.*

We are now ready to establish the main optimality result for the fully observed setting.

Theorem 3.8. *Under the conditions of Theorem 3.5, with probability at least $1 - O(n^{-2})$, the limit point G^∞ is the unique global maximizer of* (OTSM)*, up to a common rotation, and satisfies*

$$d_F(G^\infty, G^\star) \lesssim \sigma, \qquad d_\infty(G^\infty, G^\star) \lesssim \sigma\sqrt{(\log n)/n}.$$

Moreover, $G^\infty(G^\infty)^\top$ is the unique global maximizer of the semidefinite relaxation.

3.2 Random Observation Model

In this section, we consider the random observation model, in which each block on and above the diagonal of the information matrix C in the fully observed setting is independently observed with probability p.

Let $E = [e_{ij}]_{(i,j)\in[n]\times[n]}$ be a symmetric random matrix, where e_{ij} are Bernoulli random variables satisfying $\mathbb{P}(e_{ij} = 1) = p$ and $\mathbb{P}(e_{ij} = 0) = 1 - p$. The entries of E on and above the diagonal are mutually independent, and E is independent of the noise matrix W. Define

$$(G^\star G^{\star\top})_\star := [\, e_{ij}G_i^\star G_j^{\star\top}\,]_{(i,j)\in[n]\times[n]}, \quad W_\star := [\, e_{ij}W_{ij}\,]_{(i,j)\in[n]\times[n]}.$$

Since the observation pattern follows an Erdős–Rényi model, we have $\mathbb{E}\big[(G^\star G^{\star\top})_\star\big] = p\,G^\star G^{\star\top}$. Define the fluctuation term

$$\Delta_\star := (G^\star G^{\star\top})_\star - p\,G^\star G^{\star\top}.$$

Then the information matrix can be decomposed as

$$C_\star = p\,G^\star G^{\star\top} + \Delta_\star + \sigma W_\star. \tag{13}$$

Next, define the rescaled matrix $C_p := C_\star/p$, namely,

$$C_p = G^\star G^{\star\top} + \frac{1}{p}\Delta_\star + \frac{\sigma}{p}W_\star. \tag{14}$$

This formulation has the same signal term as the fully observed model, with rescaled effective noise $\widetilde{W} := \Delta_\star/p + \sigma W_\star/p$. The rescaled effective noise remains independent across unordered block pairs, and every scalar entry has a sub-Gaussian tail, although entries within one block share the same Bernoulli mask and may be dependent. In the analysis of the fully observed case, all required properties of the noise are encapsulated in Lemma C.1 and Lemma C.2. Since analogous bounds are available in the random observation setting, as established in Lemma D.1 and Lemma D.2, the arguments developed for the fully observed model extend directly to the random observation model.

Theorem 3.9. *Assume that $p \gtrsim \frac{\log n}{\sqrt{n}}$ and $\sigma \lesssim \sqrt{np/\log n}$. Then, with probability at least $1 - O(n^{-2})$, the spectrally initialized generalized power method converges linearly to the unique globally optimal solution of* (OTSM)*, up to a common rotation, and the semidefinite relaxation is tight. Moreover, the limiting solution G^∞ satisfies*

$$d_F(G^\infty, G^\star) \lesssim \frac{1+\sigma}{\sqrt{p}}, \quad d_\infty(G^\infty, G^\star) \lesssim (1+\sigma)\sqrt{\frac{\log n}{np}}.$$

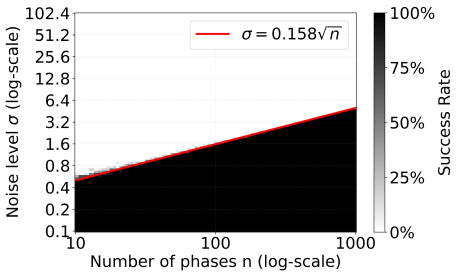


Figure 1: Fully observed model: success rate of GPM converging to the global optimum of OTSM.

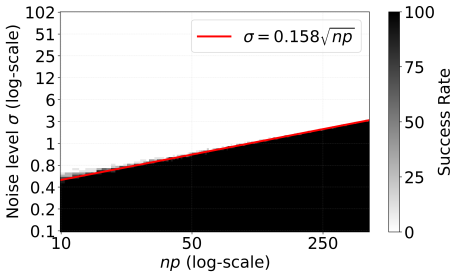


Figure 2: Random observation model: success rate of GPM converging to the global optimum of OTSM.

4 Experimental Results

4.1 Verifying Convergence of GPM

In this section, we conduct numerical experiments to verify that the generalized power method (GPM) converges to the global optimum of the OTSM problem under the noise regimes predicted by our theory.

Fully observed model. We generate 100 values of the noise level σ ranging from 0.1 to 100, together with 200 values of n ranging from 10 to 1000. For each pair (n, σ), we generate 100 independent OTSM instances with dimensions $d = 10$ and $r = 3$ according to the Gaussian noise model (5). Each instance is solved using the GPM (Algorithm 1). The algorithm is terminated when the Frobenius norm of the difference between two consecutive iterates falls below 10^{-8}, or when the maximum number of iterations reaches 400.

After obtaining the GPM estimator $\widehat{G}^{\infty}$, we verify its global optimality using the dual certificate in Lemma 3.7. Specifically, a solution $\widehat{G}^{\infty}$ is declared optimal if the corresponding matrix $L = L(\widehat{G}^{\infty}, \Lambda)$ is positive semidefinite. Due to numerical precision, we treat L as positive semidefinite if $\lambda_{\min}(L) \geq -10^{-5}|\lambda_{\max}(L)|$, which is consistent with the criterion in [6]. For each pair (n, σ), the success rate is computed as the fraction of instances (out of 100) for which GPM converges to a certified global optimum. The results are shown in Figure 1.

In Figure 1, the horizontal axis corresponds to n and the vertical axis to σ, both displayed on logarithmic scales. The shading at each (n, σ) indicates the success rate, with black representing a success rate of 1 and white representing 0. The red reference line corresponds to $\sigma = 0.158\sqrt{n}$. The figure shows that GPM successfully converges to the global optimum as long as the noise level remains below approximately $0.158\sqrt{n}$, which agrees with our theoretical noise scaling up to a logarithmic factor.

Random observation model. We next consider the random observation setting and fix the observation probability to be

$$p = 2\log n/\sqrt{n},$$

and repeat the above experiment using the same ranges of n and σ. For visualization, we plot the success rate with np on the horizontal axis and σ on the vertical axis. The results are shown in Figure 2.

As shown in Figure 2, GPM successfully converges to the global optimum until the noise level reaches approximately $0.158\sqrt{np}$, again agreeing with our theoretical noise scaling up to a logarithmic factor.

4.2 Verifying the Bound $d_F(G^{\infty}, G^{\star}) \lesssim (1+\sigma)/\sqrt{p}$

We now empirically verify the error bound established in Theorem 3.9. We generate 100 values of n ranging from 10 to 1000. For each n, we generate 10 random observation OTSM instances with parameters $d = 10$, $r = 3$, observation probability $p = 2\log n/\sqrt{n}$ and noise level $\sigma = 0.15\sqrt{np}$.

We apply GPM to obtain the estimator $\widehat{G}^{\infty}$ and compute the average recovery error $d_F(\widehat{G}^{\infty}, G^{\star})$ over the 10 instances. For comparison, we also compute the average error of the spectral ini-

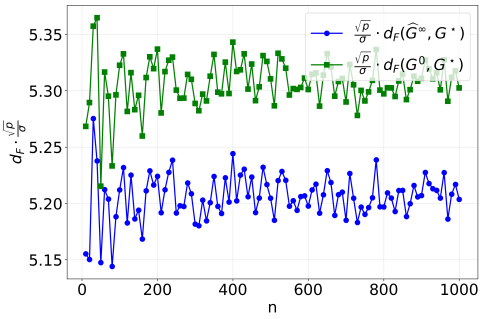


Figure 3: $d_F(\widehat{G}^{\infty}, G^{\star})$vs $d_F(G^0, G^{\star})$.

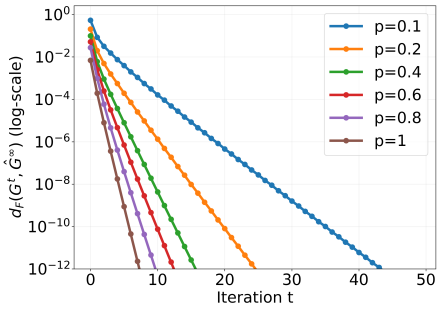


Figure 4: Linear convergence of GPM

tialization $d_F(G^0, G^{\star})$. To illustrate the scaling behavior, we plot n on the horizontal axis and $\frac{\sqrt{p}}{\sigma} d_F(\widehat{G}^{\infty}, G^{\star})$ together with $\frac{\sqrt{p}}{\sigma} d_F(G^0, G^{\star})$ on the vertical axis. The results are shown in Figure 3.

From Figure 3, we observe that as n varies from 10 to 1000, both $\frac{\sqrt{p}}{\sigma} d_F(\widehat{G}^{\infty}, G^{\star})$ and $\frac{\sqrt{p}}{\sigma} d_F(G^0, G^{\star})$ remain essentially constant, thereby validating the predicted $O((1+\sigma)/\sqrt{p})$ error bound. Moreover, the results clearly indicate that the GPM estimator $\widehat{G}^{\infty}$ achieves a consistent improvement over the spectral initialization G^0.

The improvement is modest under the relatively large sampling probability $p = 2\log n/\sqrt{n}$, which is chosen to satisfy our theoretical assumptions. Under a smaller sampling probability $p = 0.05\log n/\sqrt{n}$, GPM yields a more pronounced improvement of over 10% compared with spectral initialization; see Figure 5 in Appendices.

4.3 Verifying Linear Convergence Rate of GPM.

To verify the linear convergence of GPM, we generate OTSM instances with $n = 100$, $d = 10$, $r = 3$, and $p \in \{0.1, 0.2, 0.4, 0.6, 0.8, 1\}$. For each instance, we run GPM for 400 iterations and use G^{400} as an approximation of the limit point $\widehat{G}^{\infty}$. We then plot $d_F(G^t, \widehat{G}^{\infty})$ for the first 50 iterations in Figure 4. The nearly linear decay of $\log d_F(G^t, \widehat{G}^{\infty})$ with respect to t provides clear empirical evidence of linear convergence.

5 Conclusion and Limitation

This paper establishes algorithmic guarantees for orthogonal trace–sum maximization in a noise regime that is nearly information-theoretically optimal for statistical recovery. Using the leave-one-out technique, we prove global linear convergence of the generalized power method and tightness of the semidefinite relaxation under sharp noise conditions, for both fully observed and randomly observed settings.

The main limitations of this work are the generative model assumption (1) and the assumption of independent sub-Gaussian noise. Our analysis does not cover the most general OTSM problem, nor does it cover dependent or heavy-tailed noise.

Acknowledgments and Disclosures

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Appendices

A Related Work

Synchronization. The complex-valued synchronization problem corresponding to (OTSM) with $d_i = r = 1$ is commonly referred to as phase synchronization [2, 6, 22, 31]. For this setting, both tightness of the semidefinite relaxation and linear convergence of the generalized power method to the global optimum were established in [31] under the nearly optimal noise regime $\sigma = O(\sqrt{n/\log n})$. The real-valued counterpart with $d_i = r \geq 2$ is known as the orthogonal group synchronization problem [3, 23, 19]. In this case, [19] proved linear convergence of GPM and tightness of the SDP relaxation under the same noise condition $\sigma = O(\sqrt{n/\log n})$.

Leave-One-Out Technique. The leave-one-out (LOO) technique is a validation and sensitivity analysis method widely used in statistics, machine learning, and optimization to assess the dependence of an estimator or algorithm on individual data samples. It was first introduced in [31] to analyze the convergence behavior of the generalized power method for phase synchronization, leading to nearly optimal theoretical guarantees. Since then, the LOO technique has been successfully applied to a variety of problems to establish sharp statistical and algorithmic bounds, including entrywise eigenvector analysis [1], top-K ranking [7], stochastic block models [9], orthogonal group synchronization [20], and robust orthogonal synchronization [21].

Relation to prior work. The statement that prior GPM results guarantee only stationary-point convergence concerns general rectangular OTSM, not orthogonal-group synchronization. Table 1 compares GPM convergence to a global optimum and SDR tightness.

Compared with Zhu–Wang–So [32], our partial-observation threshold is larger by $n^{1/4}/\sqrt{\log n}$ for fixed dimensions. We also establish SDR tightness and allow heterogeneous rectangular factors, while [32] permits sparser graphs. Ling [19] is the closest square case: we match its fixed-dimensional full-observation order and extend the joint GPM–SDR theory to heterogeneous rectangular factors, sub-Gaussian noise, and missing observations. For general Stiefel OTSM, Won–Zhang–Zhou [29] prove SDR tightness at $O(n^{1/4})$; we improve this to $O(\sqrt{n/\log n})$, prove global linear convergence of GPM, and cover random observations. The rectangular extension additionally requires perturbation analysis of the rectangular polar projection and the certificate term $\tau_i(I_{d_i} - O_i O_i^\top)$, which vanishes in the square case.

Table 1: Noise upper bounds for linear convergence of GPM and SDR tightness. Noise entries have constant scale before multiplication by σ; bounds suppress absolute constants. A checkmark denotes a guarantee of linear GPM convergence to a global optimum or SDR tightness, respectively, and "–" denotes no such guarantee in the cited result.

Reference	Factor set	Noise	Sampling	Full σ	Partial σ	GPM converge	SDR Tight
Zhu–Wang–So [32]	$O(d)$	Gaussian	$p \gtrsim d/\sqrt{n}$	$n^{1/4}/d$	$n^{1/4}\sqrt{p}/d$	✓	–
Ling (GPM) [19]	$O(d)$	Gaussian	$p = 1$	$\frac{\sqrt{n}}{\sqrt{d}(\sqrt{d}+\sqrt{\log n})}$	–	✓	✓
Won–Zhang–Zhou [29]	$O_{d_i,r}$	Gaussian	$p = 1$	$n^{1/4}$	–	–	✓
This work	$O_{d_i,r}$	Sub-Gaussian	$p \gtrsim \log n/\sqrt{n}$	$\sqrt{\frac{n}{\log n}}$	$\sqrt{\frac{np}{\log n}}$	✓	✓

Here d is the common block dimension in the square cases. The bounds of [29] and this work are displayed for fixed block dimensions and rank. The GPM guarantees use spectral initialization, the prescribed step sizes, and the sampling regimes shown above.

Existing minimax lower bounds concern phase synchronization and square orthogonal/rotation synchronization [10, 11]; to our knowledge, a corresponding bound for general heterogeneous rectangular Stiefel factors under (1) has not been available. Table 2 compares these statistical lower bounds with Theorem 3.2. For normalized quotient squared loss, the phase and orthogonal-group lower bounds are asymptotically $\sigma^2/(2np)$ and $\sigma^2 d(d-1)/(2np)$, respectively, under their Gaussian normalizations and high-signal regimes. Our lower bound adds the rectangular column-space contribution $n^{-1}\sum_i r(d_i - r)$ to the rotational contribution $r(r-1)/2$, yielding the same statistical noise scale for nondegenerate fixed-dimensional factors.

Table 2: Information-theoretic minimax lower bounds for normalized quotient squared loss under partial observation, up to universal constant factors.

Reference	Factor set	Noise	Minimax lower bound
Gao–Zhang (phase) [10]	$U(1) = \{z \in \mathbb{C} : \lvert z\rvert = 1\}$	Complex Gaussian	$\dfrac{\sigma^2}{np}$
Gao–Zhang (orthogonal) [11]	$O(d), SO(d)$, $d \geq 2$	Gaussian	$\dfrac{\sigma^2 d(d-1)}{np}$
This work	$O_{d_i,r}$	Gaussian	$\dfrac{\sigma^2}{np}\left(\dfrac{1}{n}\sum_{i=1}^{n} r(d_i - r) + \dfrac{r(r-1)}{2}\right)$

All rows use independent Bernoulli(p) observations and squared quotient loss normalized by n, with the estimator class and Gaussian normalization of the respective result. Universal prefactors are suppressed. The results in [10, Theorem 5] and [11, Theorems 3.1–3.2] hold asymptotically as $n \to \infty$ and $\sigma^2/(np) \to 0$; the orthogonal/rotation result assumes fixed $d \geq 2$. Our row is the right-hand side of (4) with its universal prefactor suppressed, valid when $\sigma^2 \max\{h_{\max}, q\} \leq c_0 np$, where $h_{\max} = \max_i r(d_i - r)$ and $q = r(r-1)/2$.

B Supplementary Experiments

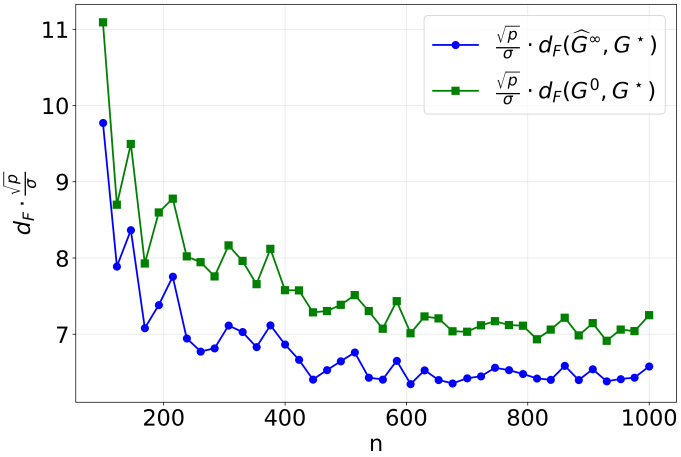


Figure 5: Comparison of $\frac{\sqrt{p}}{\sigma} d_F(\widehat{G}^{\infty}, G^{\star})$ and $\frac{\sqrt{p}}{\sigma} d_F(G^0, G^{\star})$ under a smaller sampling probability $p = 0.05 \log n/\sqrt{n}$ and noise level $\sigma = 0.15\sqrt{np}$.

C Proof of the full observation model

C.1 Noise estimates for the full observation model

We collect the operator-norm and removed-block concentration estimates used in the fully observed analysis.

Lemma C.1. *With probability at least $1 - O(n^{-2})$, the following bounds hold simultaneously:*

$$\|W\|_2 \leq C_2\sqrt{n}, \quad \|W^{(m)}\|_2 \leq C_2\sqrt{n}, \quad \|\Delta W^{(m)}\|_2 \leq C_2\sqrt{n}, \quad \|w_m\|_2 \leq C_2\sqrt{n}, \tag{15}$$

where $C_2 > 0$ is an absolute constant.

Proof. By [28, Corollary 4.4.8], we have that with probability at least $1 - 4exp(-t^2)$,

$$\|W\|_2 \leq C(\sqrt{nd} + t). \tag{16}$$

Taking $t = \sqrt{nd}$, we get the desired result. For $W^{(m)}$ and $\Delta W^{(m)}$ we can prove the bounds similarly.

For w_m, $\forall x \in \mathbb{R}^{nd}$, $\|x\|_2 = 1$

$$\|w_m^\top x\|_2 \leq \|Wx\|_2 \leq \|W\|_2, \tag{17}$$

where the first inequality is because all elements of $w_m^\top x$ are a part of vector Wx. Therefore $\|w_m\|_2 = \|w_m^\top\|_2 \leq \|W\|_2$. □

To proceed, define $U^{(m)} \subseteq \mathbb{R}^{nd\times r}$, $1 \leq m \leq n$, be collections of random matrices satisfying, for all $u \in U^{(m)}$, $\|u\|_F = \sqrt{nr}$ and u is independent of $\Delta W^{(m)}$. Moreover, assume that for each m, the cardinality of $U^{(m)}$ is deterministic and bounded by $3n^2$.

A key ingredient of our analysis is a sharp bound on the quantity $\max_{u\in U^{(m)}} \|\Delta W^{(m)}u\|_F$ where we take either $U^{(m)} = \{\widetilde{G}^{(m)}\}$ or $U^{(m)} = \{G^{t,m} : 0 \leq t \leq 3n^2 - 1\}$, which allows us to obtain an $O(\sqrt{n\log n})$ upper bound based on the following lemma.

Lemma C.2. *With probability at least $1 - O(n^{-2})$, the following holds for all $1 \leq m \leq n$:*

$$\max_{u\in U^{(m)}} \|\Delta W^{(m)}u\|_F \leq C_1\sqrt{n\log n} + C_1\sqrt{n}\, M_m, \tag{18}$$

where $C_1 > 0$ is an absolute constant and $M_m := \max_{u\in U^{(m)}} \|u_m\|_F$, with $u_m \in \mathbb{R}^{d_m\times r}$ denoting the m-th block of u.

Proof of Lemma C.2. We only need to prove that for a fixed m and a non-random $U^{(m)}$, with probability at least $1 - O(n^{-3})$, $\max_{u\in U^{(m)}} \|\Delta W^{(m)}u\|_F \leq C_1\sqrt{n\log n} + C_1\sqrt{n}M_m$. The reasons are as follows. First, if we can prove the inequality for non-random $U^{(m)}$, then leveraging independence of $\Delta W^{(m)}$ and $U^{(m)}$, we can prove for random $U^{(m)}$ by taking conditional probability and using law of Total Probability. Second, if we can prove the inequality for fixed m with probability at least $1 - O(n^{-3})$, then by taking a union bound over $m \in [n]$, we can derive inequality (18) with probability at least $1 - O(n^{-2})$.

Now, we prove that for a fixed m and a non-random $U^{(m)}$, with probability at least $1 - O(n^{-3})$, $\max_{u\in U^{(m)}} \|\Delta W^{(m)}u\|_F \leq C_1\sqrt{n\log n} + C_1\sqrt{n}M_m$.

let $u \in U^{(m)}$, we have:

$$\|\Delta W^{(m)}u\|_F^2 = \|(\Delta W^{(m)}u)_m\|_F^2 + \sum_{k\neq m} \|(\Delta W^{(m)}u)_k\|_F^2. \tag{19}$$

We control the first term and the second in the right hand of the equation respectively.

Step 1: For the first term:

$$(\Delta W^{(m)}u)_m = \sum_{k=1}^{n} W_{mk}u_k.$$

Express the (i, j) element of $(\Delta W^{(m)}u)_m$ in the following form:

$$(\Delta W^{(m)}u)_m^{ij} = \sum_{k=1}^{n} (W_{mk}u_k)^{ij} = \sum_{k=1}^{n}\sum_{s=1}^{d_k} W_{mk}^{is}u_k^{sj}.$$

For a fixed pair of (i,j), W_{mk}^{is} on the right hand side of the equality are a row of W. Therefore, these W_{mk}^{is} are mean zero , independent, standard sub-Gaussian random variables. By the sub-Gaussian Hoeffding inequality [28, Theorem 2.6.2], we have

$$P(|\sum_{k=1}^{n}\sum_{s=1}^{d_k} W_{mk}^{is}u_k^{sj}| \geq t) \leq 2exp(\frac{-ct^2}{\sum_{k=1}^{n}\sum_{s=1}^{d_k}(u_k^{sj})^2}) \leq 2exp(\frac{-ct^2}{\|u\|_F^2}) = 2exp(\frac{-ct^2}{nr}),$$

where c is an absolute constant. Let $t = C_1\sqrt{n\log n}$ with $cC_1^2 \geq 5r$, we deduce that with probability at least $1-2n^{-5}$,

$$|\sum_{k=1}^{n}\sum_{s=1}^{d_k} W_{mk}^{is}u_k^{sj}| \leq C_1\sqrt{n\log n}.$$

By taking a union bound over (i,j), we deduce that with probability at least $1-2d_mrn^{-5}$,

$$\forall (i,j) \in [d_m]\times[r], \quad |(\Delta W^{(m)}u)_m^{ij}| \leq C_1\sqrt{n\log n}.$$

As a result, with probability at least $1-2d_mrn^{-5}$,

$$\|(\Delta W^{(m)}u)_m\|_F = \sqrt{\sum_{i=1}^{d_m}\sum_{j=1}^{r}((\Delta W^{(m)}u)_m^{ij})^2} \leq \sqrt{d_mr}C_1\sqrt{n\log n}. \tag{20}$$

Step 2: For the second term:

$$\begin{aligned}\sum_{k\neq m}\|(\Delta W^{(m)}u)_k\|_F^2 &= \sum_{k\neq m}\|W_{km}u_m\|_F^2\\ &\leq \|u_m\|_F^2\sum_{k\neq m}\|W_{km}\|_F^2\\ &\leq M_m^2\sum_{k\neq m}\|W_{km}\|_F^2\\ &= M_m^2\sum_{k\neq m}\sum_{(i,j)\in[d_k]\times[d_m]}(W_{km}^{ij})^2.\end{aligned} \tag{21}$$

W_{km}^{ij} on the right hand of the last equality are $(nd-d_m)d_m$ mean zero, independent, standard sub-Gaussian random variables. Therefore, $(W_{km}^{ij})^2$ are $(nd-d_m)d_m$ independent sub-exponential random variables. Let $\mu_{km}^{ij} = \mathbb{E}[(W_{km}^{ij})^2]$, by Bernstein's inequality [28, Theorem 2.8.1]:

$$P(|\sum_{k\neq m}\sum_{(i,j)\in[d_k]\times[d_m]}[(W_{km}^{ij})^2 - \mu_{km}^{ij}]| \geq t) \leq 2exp(-c\min(\frac{t^2}{(nd-d_m)d_m}, t)),$$

where c is an absolute constant. Taking $t = cndd_m$, we derive that with probability at least $1-2exp(-cndd_m)$,

$$|\sum_{k\neq m}\sum_{(i,j)\in[d_k]\times[d_m]}[(W_{km}^{ij})^2 - \mu_{km}^{ij}]| \leq ndd_m.$$

Since $\|W_{km}^{ij}\|_{\psi_2} \leq 1$, the moment bound in [28, Proposition 2.5.2] gives $\mu_{km}^{ij} \leq \mu' := 2(C')^2$ uniformly. With probability at least $1-2exp(-cndd_m)$,

$$\begin{aligned}\sum_{k\neq m}\sum_{(i,j)\in[d_k]\times[d_m]}(W_{km}^{ij})^2 &\leq \left|\sum_{k\neq m}\sum_{(i,j)\in[d_k]\times[d_m]}[(W_{km}^{ij})^2 - \mu_{km}^{ij}]\right|\\ &\quad + (nd-d_m)d_m\mu'\\ &\leq (\mu'+1)ndd_m.\end{aligned} \tag{22}$$

Combining (19),(20),(21) and (22), we derive that with probability at least $1-2exp(-cndd_m) - 2d_mrn^{-5}$,

$$\begin{aligned}\|\Delta W^{(m)}u\|_F &\leq \|(\Delta W^{(m)}u)_m\|_F + \sqrt{\sum_{k\neq m}\|(\Delta W^{(m)}u)_k\|_F^2}\\ &\leq \sqrt{d_mr}C_1\sqrt{n\log n} + \sqrt{\mu'+1}\sqrt{ndd_m}M_m\end{aligned}$$

Finally, take a union bound over $u \in U^{(m)}$ and rewrite the constant as $C_1 = \max\{\sqrt{d_{max}r}C_1, \sqrt{(\mu'+1)dd_{max}}\}$, where $d_{max} = \max_{i\in[n]} d_i$. With probability at least $1 - 6n^2 exp(-cndd_m) - 6d_m r n^{-3}$:

$$\|\Delta W^{(m)} u\|_F \leq C_1\sqrt{n\log n} + C_1\sqrt{n}M_m,$$

which completes the proof. □

C.2 Proof of Theorem 3.3

Proof of Theorem 3.3 . We claim that the top r eigenvalues of $G^\star G^{\star\top}$ are all n and other eigenvalues of $G^\star G^{\star\top}$ are 0; moreover, columns of $G^\star$ are r linearly independent eigenvectors of $G^\star G^{\star\top}$ corresponding to eigenvalue n.

Now, we prove the claim. Since $G^\star \in O^n_{d,r}$, we have

$$G^\star G^{\star\top} G^\star = G^\star n I_r = nG^\star.$$

This implies that columns of $G^\star$ are eigenvectors of $G^\star G^{\star\top}$ corresponding to the eigenvalue n. By the rank inequality $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$, we have

$$r = \text{rank}(G^{\star\top} G^\star) \leq \text{rank}(G^\star) \leq r.$$

This implies that the columns of $G^\star$ are linearly independent and the algebraic multiplicity of n is at least r. By the rank inequality $\text{rank}(AB) \geq \text{rank}(A) + \text{rank}(B) - r$ we derive,

$$r = \text{rank}(G^\star) + \text{rank}(G^{\star\top}) - r \leq \text{rank}(G^\star G^{\star\top}) \leq \text{rank}(G^\star) = r$$

This implies $\text{rank}(G^\star G^{\star\top}) = r$, so $G^\star G^{\star\top}$ has at most r non-zero eigenvalues. Combining the above discussion, the r non-zero eigenvalues must be n. This finishes the proof of the claim. Next, we prove the theorem in two steps.

Step 1: Prove $d_F(\widetilde{G}, G^\star) \leq C_0\sigma$, using standard matrix permutation bounds.

With probability at least $1 - O(n^{-2})$, the bounds in lemma C.1 hold. We choose c_0 such that $c_0 C_2 < \frac{1}{8}$. Using Davis–Kahan inequality [8] we get

$$\begin{aligned} d_F(\sqrt{d}G^\star, \sqrt{d}\widetilde{G}) &\leq \frac{\sqrt{2}\|\sigma W\sqrt{d}G^\star\|_F}{n - \|\sigma W\|_2} \\ &\leq \frac{\sqrt{2d}\sigma\|W\|_2\|G^\star\|_F}{n - \sigma\|W\|_2} \\ &\leq \frac{\sqrt{2d}\sigma C_2\sqrt{n}\sqrt{nr}}{n - c_0\sqrt{n}C_2\sqrt{n}} \\ &= \frac{8\sqrt{2dr}C_2\sigma}{7}. \end{aligned}$$

Dividing $\sqrt{d}$ on both sides, we get

$$d_F(G^\star, \widetilde{G}) \leq \frac{8\sqrt{2r}C_2\sigma}{7}$$

Step 2: Prove $d_\infty(\widetilde{G}, G^\star) \leq C_0\sigma\sqrt{\log n/n}$, using leave-one-out technique.

Let $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r$ be the largest r eigenvalues of C and let $\Lambda = diag(\lambda_1, \cdots, \lambda_r)$. Using Weyl inequality [5, Chapter III], we get

$$\begin{aligned} |\lambda_i - n| \leq \sigma\|W\|_2 \leq c_0\sqrt{n}C_2\sqrt{n} \leq \frac{1}{8}n \\ \frac{7}{8}n \leq \lambda_i \leq \frac{9}{8}n, \quad \forall i \in [r]. \end{aligned}$$

Therefore Λ is invertible. Since columns of $\widetilde{G}$ are eigenvectors of C, we have

$$\widetilde{G}\Lambda = C\widetilde{G} = G^\star G^{\star\top}\widetilde{G} + \sigma W\widetilde{G},$$
$$\widetilde{G} = G^\star G^{\star\top}\widetilde{G}\Lambda^{-1} + \sigma W\widetilde{G}\Lambda^{-1}.$$

Assume $d_F(G^\star, \widetilde{G}) = \|\widetilde{G} - G^\star Q\|_F, \quad Q \in O(r)$, Then, $\forall m \in [n]$,

$$\begin{aligned}\|(\widetilde{G} - G^\star Q)_m\|_F &= \|G_m(G^{\star\top}\widetilde{G}\Lambda^{-1} - Q) + \sigma(W\widetilde{G})_m\Lambda^{-1}\|_F \\ &\le \|G^\star_m\|_F\|G^{\star\top}\widetilde{G} - Q\Lambda\|_2\|\Lambda^{-1}\|_2 + \sigma\|(W\widetilde{G})_m\|_F\|\Lambda^{-1}\|_2 \\ &\le \frac{\sqrt{r}\|G^{\star\top}\widetilde{G} - Q\Lambda\|_2}{\frac{7}{8}n} + \frac{\sigma\|W\widetilde{G}\|_\infty}{\frac{7}{8}n}.\end{aligned} \tag{23}$$

We bound the two terms in the last inequality respectively.

We first bound $\|G^{\star\top}\widetilde{G} - Q\Lambda\|_2$.

$$\begin{aligned}\|G^{\star\top}\widetilde{G} - Q\Lambda\|_2 &\le \|G^{\star\top}\widetilde{G} - nQ\|_F + \|nQ - Q\Lambda\|_2 \\ &= \|G^{\star\top}(\widetilde{G} - G^\star Q)\|_F + \|nI_r - \Lambda\|_2 \\ &\le \sqrt{nr}d_F(\widetilde{G}, G^\star) + C_2\sigma\sqrt{n} \\ &\le \frac{8\sqrt{2}rC_2\sigma\sqrt{n}}{7} + C_2\sigma\sqrt{n} \\ &= (\frac{8\sqrt{2}rC_2}{7} + C_2)\sigma\sqrt{n}.\end{aligned} \tag{24}$$

Next, We bound $\|W\widetilde{G}\|_\infty$.

$$\|W\widetilde{G}\|_\infty = \max_{m\in[n]}\|w_m^\top\widetilde{G}\|_F, \tag{25}$$

where $w_m \in \mathbb{R}^{nd\times d_i}$ is the m-th block column of W. Take an arbitrary $m \in [n]$. Let $\widetilde{G}^{(m)}$ be a $nd \times r$ matrix, whose columns are the top r leading eigenvectors of $C^{(m)}$, with $\|\widetilde{G}^{(m)}\|_F = \sqrt{nr}$. Assuming that $d_F(\widetilde{G}, \widetilde{G}^{(m)}) = \|\widetilde{G} - \widetilde{G}^{(m)}Q\|_F,\ Q \in O(r)$, we have

$$\begin{aligned}\|w_m^\top\widetilde{G}\|_F &\le \|w_m^\top\widetilde{G}^{(m)}Q\|_F + \|w_m^\top(\widetilde{G} - \widetilde{G}^{(m)}Q)\|_F \\ &\le \|w_m^\top\widetilde{G}^{(m)}\|_F + \|w_m\|_2 d_F(\widetilde{G}^{(m)}, \widetilde{G}).\end{aligned} \tag{26}$$

Using Weyl's inequality [5, Chapter III], we have

$$|\lambda_r(C^{(m)}) - n| \le \sigma\|W^{(m)}\|_2 \le c_0C_2n \le \frac{1}{8}n,$$
$$|\lambda_{r+1}(C^{(m)}) - 0| \le \sigma\|W^{(m)}\|_2 \le c_0C_2n \le \frac{1}{8}n.$$

Therefore $\delta(C^{(m)}) = \lambda_r(C^{(m)}) - \lambda_{r+1}(C^{(m)}) \ge \frac{7}{8}n - \frac{1}{8}n = \frac{3}{4}n$. Using Davis–Kahan inequality [8], we deduce

$$\begin{aligned}d_F(\sqrt{d}\widetilde{G}^{(m)}, \sqrt{d}\widetilde{G}) &\le \frac{\sqrt{2}\|\sigma\Delta W^{(m)}\sqrt{d}\widetilde{G}^{(m)}\|_F}{\delta(C^{(m)}) - \|\sigma\Delta W^{(m)}\|_2} \\ &\le \frac{\sqrt{2d}\sigma\|\Delta W^{(m)}\widetilde{G}^{(m)}\|_F}{\frac{3}{4}n - \frac{1}{8}n} \\ &= \frac{8\sqrt{2d}\sigma\|\Delta W^{(m)}\widetilde{G}^{(m)}\|_F}{5n}.\end{aligned} \tag{27}$$

Combining (27) and (26) we have

$$\begin{aligned}\|w_m^\top\widetilde{G}\|_F &\le \|w_m^\top\widetilde{G}^{(m)}\|_F + C_2\sqrt{n}\frac{8\sqrt{2}\sigma\|\Delta W^{(m)}\widetilde{G}^{(m)}\|_F}{5n} \\ &\le \|\Delta W^{(m)}\widetilde{G}^{(m)}\|_F + \frac{8\sqrt{2}C_2c_0\|\Delta W^{(m)}\widetilde{G}^{(m)}\|_F}{5} \\ &\le (\frac{\sqrt{2}}{5} + 1)\|\Delta W^{(m)}\widetilde{G}^{(m)}\|_F,\end{aligned} \tag{28}$$

where the second inequality is because $w_m^\top \widetilde{G}$ is a block in $\Delta W^{(m)}\widetilde{G}^{(m)}$. Noticing that $\widetilde{G}^{(m)}$ is independent from $\Delta W^{(m)}$, we choose $U^{(m)} = \{\widetilde{G}^{(m)}\}$ in lemma C.2. With probability at least $1 - O(n^{-2})$,

$$\forall 1 \le m \le n, \quad \|\Delta W^{(m)}\widetilde{G}^{(m)}\|_F \le C_1\sqrt{n\log n} + C_1\sqrt{n}\|\widetilde{G}_m^{(m)}\|_F, \tag{29}$$

where $\widetilde{G}_m^{(m)}$ is the m-th $d \times r$ block of $\widetilde{G}^{(m)}$. Combining (25),(28) and (29) we derive

$$\|W\widetilde{G}\|_\infty \le (\frac{\sqrt{2}}{5} + 1)(C_1\sqrt{n\log n} + C_1\sqrt{n}\max_{m\in[n]}\|\widetilde{G}_m^{(m)}\|_F). \tag{30}$$

Next, we will prove that $\|\widetilde{G}_m^{(m)}\|_F \le \frac{8}{7}r$. Since the columns of $\widetilde{G}^{(m)}$ are leading eigenvectors of $C^{(m)}$, we have

$$\widetilde{G}^{(m)}\widetilde{\Lambda} = C^{(m)}\widetilde{G}^{(m)} = G^\star G^{\star\top}\widetilde{G}^{(m)} + \sigma W^{(m)}\widetilde{G}^{(m)},$$

where $\widetilde{\Lambda}$ is a $r \times r$ diagonal matrix whose diagonal entries are the top r largest eigenvalues of $C^{(m)}$. Using Weyl's inequality [5, Chapter III], we deduce that all diagonal entries of $\widetilde{\Lambda}$ are larger than $\frac{7}{8}n$. Therefore,$\forall m \in [n]$,

$$\begin{aligned}
\|\widetilde{G}_m^{(m)}\|_F &= \|\left(G_m^\star G^{\star\top}\widetilde{G}^{(m)} + \sigma(W^{(m)}\widetilde{G}^{(m)})_m\right)\widetilde{\Lambda}^{-1}\|_F \\
&= \|G_m^\star G^{\star\top}\widetilde{G}^{(m)}\widetilde{\Lambda}^{-1}\|_F \\
&\le \|G_m^\star\|_2\|G^{\star\top}\|_F\|\widetilde{G}^{(m)}\|_F\|\widetilde{\Lambda}^{-1}\|_2 \\
&\le \frac{8}{7}r,
\end{aligned} \tag{31}$$

where the second equality is because that the m-th block row of $W^{(m)}$ is 0. Combining (30) and (31), we get

$$\|W\widetilde{G}\|_\infty \le (\frac{\sqrt{2}}{5} + 1)(C_1\sqrt{n\log n} + \frac{8}{7}r^{\frac{3}{2}}C_1\sqrt{n}). \tag{32}$$

Finally, combining (23), (24) and (32) we get

$$\|(\widetilde{G} - G^\star Q)_m\|_F \le \frac{\sqrt{r}(\frac{8\sqrt{2}rC_2}{7} + C_2)\sigma\sqrt{n}}{\frac{7}{8}n} + \frac{\sigma(\frac{\sqrt{2}}{5} + 1)(C_1\sqrt{n\log n} + \frac{8}{7}r^{\frac{3}{2}}C_1\sqrt{n})}{\frac{7}{8}n}, \tag{33}$$

which completes the proof.

□

C.3 Local Contractivity

Lemma C.3 (Local contractivity of $\mathcal{P}_C$)**.** *Let $C = G^\star G^{\star\top} + \sigma W$ and $C' = G^\star G^{\star\top} + \sigma W'$. For $x, y \in O_{d,r}^n$, suppose there exist positive constants $k_1, k_2, \epsilon_1, \epsilon_2$ such that*

$$\begin{aligned}
d_F(x, G^\star) \le \epsilon_1\sqrt{nr}, \quad \|Wx\|_\infty \le k_1\sqrt{n\log n}, \\
d_F(y, G^\star) \le \epsilon_2\sqrt{nr}, \quad \|W'y\|_\infty \le k_2\sqrt{n\log n}.
\end{aligned} \tag{34}$$

Assume

$$\epsilon := \min\left\{\frac{\epsilon_1^2 r}{2} + \sigma k_1\sqrt{\frac{\log n}{n}},\ \frac{\epsilon_2^2 r}{2} + \sigma k_2\sqrt{\frac{\log n}{n}}\right\} < 1.$$

Then we have

$$d_F(\mathcal{P}_C(x),\, \mathcal{P}_{C'}(y)) \le \frac{4}{(1-\epsilon)n}\, d_F(Cx, C'y). \tag{35}$$

Moreover, if $C = C'$, then

$$d_F(\mathcal{P}_C(x),\, \mathcal{P}_C(y)) \le \rho\, d_F(x, y), \tag{36}$$

where

$$\rho := \frac{4}{1-\epsilon}\left(2\max\{\epsilon_1, \epsilon_2\}\sqrt{r} + \frac{\sigma\|W\|_2}{n}\right).$$

We first establish several auxiliary results on alignment and Stiefel projection, and then return to the proof of Lemma C.3.

Lemma C.4. *Let $A \in \mathbb{R}^{m\times m}$ with SVD decomposition $A = UDV^\top$ then,*

$$\|A\|_\star = \max_{Q\in O(m)} \langle Q, A\rangle = \langle UV^\top, A\rangle, \tag{37}$$

Proof. Let $Q \in O(m)$. We use $\sigma_i(A)$ to denote the i-th singular value of A and $\sigma_i(Q)$ to denote the i-th singular value of Q. Using von Neumann's inequality [5, Chapter IV], we have

$$\langle Q, A\rangle \le \sum_{i=1}^{n} \sigma_i(Q)\sigma_i(A) \le \sum_{i=1}^{n} \sigma_i(A) = \|A\|_\star, \tag{38}$$

where the second inequality is because $|\sigma_i(Q)| = 1$. On the other hand, let $Q = UV^\top$, we have

$$\langle Q, A\rangle = tr(A^\top Q) = tr(VDV^\top) = tr(D) = \|A\|_\star. \tag{39}$$

Therefore (37) holds. □

Proposition C.5. *Let $G_1, G_2 \in O^n_{d,r}$ with $d_F(G_1, G_2) \le \epsilon\sqrt{nr}$, then $\forall i \in [r]$,*

$$(1 - \frac{\epsilon^2 r}{2})n \le \sigma_i(G_1^\top G_2) \le n. \tag{40}$$

Proof. Since $d_F(G_1, G_2) \le \epsilon\sqrt{nr}$, we have

$$\begin{aligned} \epsilon^2 nr \ge d_F^2(G_1, G_2) &= \|G_1\|_F^2 + \|G_2\|_F^2 - 2\|G_1^\top G_2\|_\star \\ &= 2\sum_{i=1}^{r}(n - \sigma_i(G_1^\top G_2)) \ge 2(n - \sigma_r(G_1^\top G_2)), \end{aligned} \tag{41}$$

which implies

$$\sigma_r(G_1^\top G_2) \ge (1 - \frac{\epsilon^2 r}{2})n. \tag{42}$$

On the other hand, $\forall i \in [r]$,

$$\sigma_i(G_1^\top G_2) \le \|G_1^\top G_2\|_2 \le \|G_1\|_2\|G_2\|_2 = n. \tag{43}$$

□

Lemma C.6. *Let $L = \frac{1}{n}C$, $x, y \in O^n_{d,r}$, $d_F(x, G^\star) \le \epsilon\sqrt{n}$, $d_F(y, G^\star) \le \epsilon\sqrt{n}$, then*

$$d_F(Lx, Ly) \le (2\epsilon + \sigma\frac{\|W\|_2}{n})d_F(x, y) \tag{44}$$

Proof. Let $d_F(x, y) = \|x - yQ\|_F$, we claim that

$$\langle y^\top x, Q\rangle = \|y^\top x\|_\star. \tag{45}$$

The reason is as follows: $Q = argmin_{P\in O(r)}\|x - yP\|_F^2 = argmax_{P\in O(r)}\langle x, yP\rangle = argmax_{P\in O(r)}\langle y^\top x, P\rangle$. By lemma C.4, $\langle y^\top x, Q\rangle = \max_{Q\in O(r)}\langle Q, y^\top x\rangle = \|y^\top x\|_\star$.

We now prove lemma C.6.

$$\begin{aligned} d_F(Lx, Ly) &\le \|Lx - LyQ\|_F \\ &= \|\frac{1}{n}(G^\star G^{\star\top} + \sigma W)(x - yQ)\|_F \\ &\le \frac{1}{n}\|G^\star\|_2\|G^{\star\top}(x - yQ)\|_F + \frac{\sigma}{n}\|W(x - yQ)\|_F \\ &\le \frac{\|G^{\star\top}(x - yQ)\|_F}{\sqrt{n}} + \frac{\sigma\|W\|_2}{n}d_F(x, y). \end{aligned} \tag{46}$$

The second term on the right of the last inequality is exactly what we wanted and we only needs to bound the first term $\|G^{\star\top}(x - yQ)\|_F$. Letting $d_F(y, G^\star) = \|y - G^\star Q_y\|_F$, we have

$$\begin{aligned}\|G^{\star\top}(x - yQ)\|_F &\le \|(y - G^\star Q_y)^\top (x - yQ)\|_F + \|y^\top (x - yQ)\|_F \\ &\le d_F(y, G^\star) d_F(x, y) + \|y^\top (x - yQ)\|_F \\ &\le \epsilon\sqrt{n} d_F(x, y) + \|y^\top x - nQ)\|_F.\end{aligned} \tag{47}$$

Using $\sigma_i(y^\top x)$ to denote the i-th singular value of $y^\top x$, we have

$$\begin{aligned}\|y^\top x - nQ)\|_F^2 &= \|y^\top x\|_F^2 + n^2 r - 2n < Q, y^\top x > \\ &= \sum_{i=1}^r (\sigma_i(y^\top x))^2 + n^2 r - 2n \sum_{i=1}^r \sigma_i(y^\top x) \\ &= \sum_{i=1}^r (n - \sigma_i(y^\top x))^2,\end{aligned} \tag{48}$$

where the second equality comes from (45). Since $d_F(x, y) \le d_F(x, G^\star) + d_F(y, G^\star) \le 2\epsilon\sqrt{n}$, using proposition (C.5), we have

$$(1 - 2\epsilon^2)n \le \sigma_i(y^\top x) \le n. \tag{49}$$

Combining (48),(49), we have

$$\begin{aligned}\|y^\top x - nQ)\|_F^2 &= \sum_{i=1}^r (n - \sigma_i(y^\top x))^2 \\ &\le (n - \sigma_r(y^\top x)) \sum_{i=1}^r \big(n - \sigma_i(y^\top x)\big) \\ &\le 2\epsilon^2 n(nr - \|y^\top x\|_\star) \\ &= \epsilon^2 n d_F^2(x, y).\end{aligned} \tag{50}$$

Combining (46),(47), (50), we have

$$\begin{aligned}d_F(Lx, Ly) &\le \frac{\|G^{\star\top}(x - yQ)\|_F}{\sqrt{n}} + \frac{\sigma\|W\|_2}{n} d_F(x, y) \\ &\le 2\epsilon d_F(x, y) + \frac{\sigma\|W\|_2}{n} d_F(x, y) \\ &= (2\epsilon + \frac{\sigma\|W\|_2}{n}) d_F(x, y).\end{aligned} \tag{51}$$

□

Lemma C.7. *Let $X, Y \in \mathbb{R}^{d\times r}$, $\sigma_r(X) > 0$, then*

$$\|\pi(X) - \pi(Y)\|_F \le \frac{4\|X - Y\|_F}{\sigma_r(X)}, \tag{52}$$

where π is the projection operator on stiefel manifold.

Proof. Using SVD decomposition, $X = U_X D_X V_X^\top$, where $U_X \in \mathbb{R}^{d\times r}$ satisfies $U_X^\top U_X = I_r$, $D_X = diag(\sigma_1(X), \cdots \sigma_r(X))$, $V_X \in O(r)$. Similarly we have the SVD decomposition $Y = U_Y D_Y V_Y^\top$. We construct two auxiliary matrices as follows:

$$\widetilde{X} = \begin{pmatrix} O_{d\times d} & X \\ X^\top & O_{r\times r}\end{pmatrix}, \quad \widetilde{Y} = \begin{pmatrix} O_{d\times d} & Y \\ Y^\top & O_{r\times r}\end{pmatrix}.$$

Notice:

$$\begin{aligned}|\lambda I - \widetilde{X}| &= \begin{vmatrix} \lambda I_d & -X \\ -X^\top & \lambda I_r\end{vmatrix} = \begin{vmatrix} \lambda I_d & -X \\ O & \lambda I_r - \frac{1}{\lambda} X^\top X\end{vmatrix} \\ &= \lambda^{d-r} |\lambda^2 I_r - X^\top X| \\ &= \lambda^{d-r} (\lambda^2 - \sigma_1(X)^2)(\lambda^2 - \sigma_2(X)^2) \cdots (\lambda^2 - \sigma_r(X)^2).\end{aligned}$$

We deduce that the eigenvalues of $\widetilde{X}$ are $\sigma_1(X), \cdots, \sigma_r(X)$, 0 with multiplicity $d-r$, and $-\sigma_1(X), \cdots - \sigma_r(X)$. Similarly, the eigenvalues of Y are $\sigma_1(Y), \cdots, \sigma_r(Y)$, 0 with multiplicity $d-r$, and $-\sigma_1(Y), \cdots - \sigma_r(Y)$.

Define

$$M_X = \frac{1}{\sqrt{2}} \begin{pmatrix} U_X \\ V_X \end{pmatrix}, \quad M_Y = \frac{1}{\sqrt{2}} \begin{pmatrix} U_Y \\ V_Y \end{pmatrix}.$$

. It is obvious that $M_X^\top M_X = M_Y^\top M_Y = I_r$. Besides, we have $\widetilde{X} M_X = M_X D_X$ and $\widetilde{Y} M_Y = M_Y D_Y$. Thus, M_X and M_Y are the top r leading eigenvectors of $\widetilde{X}$ and $\widetilde{Y}$ respectively. Using Davis–Kahan inequality [8] we get

$$d_F(M_X, M_Y) \leq \frac{\sqrt{2}\|(\widetilde{Y} - \widetilde{X}) M_X\|_F}{\sigma_r(X)} \leq \frac{\sqrt{2}\|(\widetilde{Y} - \widetilde{X})\|_F \|M_X\|_2}{\sigma_r(X)} = \frac{2\|(Y-X)\|_F}{\sigma_r(X)}.$$

From above we deduce that there exists $Q \in O(r)$ such that

$$\| \begin{pmatrix} U_X - U_Y Q \\ V_X - V_Y Q \end{pmatrix} \|_F = \sqrt{\|U_X - U_Y Q\|_F^2 + \|V_X - V_Y Q\|_F^2} \leq \frac{2\sqrt{2}\|(Y-X)\|_F}{\sigma_r(X)}.$$

On the other hand,

$$\begin{aligned} \|\pi(X) - \pi(Y)\|_F &= \|U_X V_X^\top - U_Y V_Y^\top\|_F \\ &\leq \|U_X V_X^\top - U_Y Q V_X^\top\|_F + \|U_Y Q V_X^\top - U_Y V_Y^\top\|_F \\ &\leq \|U_X - U_Y Q\|_F + \|V_X - V_Y Q\|_F \\ &\leq \sqrt{2}\sqrt{\|U_X - U_Y Q\|_F^2 + \|V_X - V_Y Q\|_F^2} \\ &\leq \frac{4\|(Y-X)\|_F}{\sigma_r(X)}. \end{aligned}$$

□

Corollary C.8. *Let $X, Y \in \mathbb{R}^{nd \times r}$ with $\min_{i \in [n]} \sigma_r(X_i) \geq \epsilon > 0$, then*

$$d_F(\pi^n(X), \pi^n(Y)) \leq \frac{4 d_F(X,Y)}{\epsilon} \tag{53}$$

Proof. Assume that $d_F(X,Y) = \|X - YQ\|_F$.

$$\begin{aligned} d_F(\pi^n(X), \pi^n(Y)) \leq \|\pi^n(X) - \pi^n(Y)Q\|_F &= \sqrt{\sum_{i=1}^n \|\pi(X_i) - \pi(Y_i Q)\|_F^2} \\ &\leq \frac{4}{\epsilon}\sqrt{\sum_{i=1}^n \|X_i - Y_i Q\|_F^2} = \frac{4}{\epsilon} d_F(X,Y). \end{aligned}$$

□

Proof of Lemma C.3. Let $L = \frac{1}{n}C, L' = \frac{1}{n}C'$. In order to use corollary C.8 , we first provide a lower bound for $\sigma_r((Lx)_i)$ and $\sigma_r((L'y)_i)$.

Weyl's singular-value inequality [5, Chapter III] gives

$$\sigma_r(A) \geq \sigma_r(A+B) - \sigma_1(B). \tag{54}$$

Letting $A = (Lx)_i$, $B = -\frac{\sigma}{n}(Wx)_i$ and $A + B = (Lx)_i - \frac{\sigma}{n}(Wx)_i = \frac{G_i^\star G^{\star\top} x}{n}$, we derive

$$\sigma_r((Lx)_i) \geq \frac{1}{n}\sigma_r(G_i^\star G^{\star\top} x) - \frac{\sigma}{n}\sigma_1((Wx)_i) = \frac{1}{n}\sigma_r(G_i^\star G^{\star\top} x) - \frac{\sigma}{n}\|Wx\|_\infty. \tag{55}$$

Since $d_F(x, G^\star) \le \epsilon_1\sqrt{nr}$, by proposition C.5, we have $\sigma_r(G^{\star\top}x) \ge (1 - \frac{\epsilon_1^2 r}{2})n$. Therefore, $\sigma_r((Lx)_i) \ge 1 - \frac{\epsilon_1^2 r}{2} - \sigma k_1\sqrt{\log n/n}$. Similarly, we have $\sigma_r((L'y)_i) \ge 1 - \frac{\epsilon_2^2 r}{2} - \sigma k_2\sqrt{\log n/n}$. Using corollary C.8 we derive

$$d_F(\pi^n(Lx), \pi^n(Ly)) \le \frac{4}{1-\epsilon} d_F(Lx, Ly). \tag{56}$$

If $L = L'$, using lemma C.6 we have:

$$d_F(Lx, Ly) \le (2\max\{\epsilon_1, \epsilon_2\}\sqrt{r} + \frac{\sigma\|W\|_2}{n}) d_F(x, y). \tag{57}$$

Combining (56), (57), we complete the proof. □

C.4 Proof of Proposition 3.4

With probability at least $1 - O(n^{-2})$, the results in Theorem 3.3 hold.

Step 1: $d_F(G^0, G^\star) \lesssim \sigma$.

$$d_F(G^0, G^\star) = d_F(\pi^n(\widetilde{G}), \pi^n(G^\star)) \le 4 d_F(\widetilde{G}, G^\star) \le 4C_0\sigma, \tag{58}$$

where the first inequality is because of corollary C.8 and $\sigma_r(G_i^\star) = 1$.

Step 2: $d_\infty(G^0, G^\star) \lesssim \sigma\sqrt{n/\log n}$.

Assume that $d_\infty(\widetilde{G}, G^\star) = \|\widetilde{G} - G^\star Q\|_\infty$, then $\forall i$

$$\|G_i^0 - G_i^\star Q\|_F = \|\pi(\widetilde{G}_i) - \pi(G_i^\star Q)\|_F \le 4\|\widetilde{G}_i - G_i^\star Q\|_F \le 4 d_\infty(\widetilde{G}, G^\star). \tag{59}$$

Therefore, $d_\infty(G^0, G^\star) \le \|G^0 - G^\star Q\|_\infty \le 4 d_\infty(G^\star, \widetilde{G}) \lesssim \sigma\sqrt{\log n/n}$.

C.5 Proof of Theorem 3.5

We split the proof into two parts: Theorem C.9 handles the initial point $G^0, G^{0,m}$ and Theorem C.10 handles the rest of points by induction.

Theorem C.9. *Let $n \ge 2$, $\sigma \le \min\{\frac{\sqrt{n/r}}{12000\sqrt{2}C_2}, \frac{1}{24000k_2}\sqrt{\frac{n/r}{\log n}}, C_3\sqrt{n/\log n}\}$, then with probability at least $1 - O(n^{-2})$, we have*

$$d_F(G^0, G^\star) \le k_3\sqrt{n}, \tag{60}$$

$$d_F(G^{0,m}, G^0) \le k_1, \tag{61}$$

$$\|WG^0\|_\infty \le k_2\sqrt{n\log n}, \tag{62}$$

where C_2, C_1 are constants in lemma C.1 and lemma C.2 respectively. c_0 and C_0 are constants in theorem 3.3. $C_3 \le c_0$ and $C_3C_0 \le \frac{1}{2}$. $k_1 = k_3 = \frac{1}{600}$, $k_2 = 4C_1\sqrt{r} + 2C_2k_1$.

Proof. **Step 1:** $d_F(G^0, G^\star) \le k_3\sqrt{n}$.

Notice that $\min_{i\in[n]}\sigma_r(G_i^\star) = 1$. Using corollary C.8 we have:

$$d_F(G^0, G^\star) = d_F(\pi^n\widetilde{G}, \pi^n G^\star) \le 4 d_F(\widetilde{G}, G^\star). \tag{63}$$

Using Davis–Kahan inequality [8], we derive

$$d_F(\widetilde{G}, G^\star) \le \frac{\sqrt{2}\sigma\|W\|_2\|G^\star\|_F}{n - \sigma\|W\|_2} \le \frac{\sqrt{2}\sigma C_2\sqrt{n}\sqrt{nr}}{n - \sigma C_2\sqrt{n}} \le \frac{1}{6000}\sqrt{n}. \tag{64}$$

Therefore,$d_F(G^0, G^\star) \le \frac{4}{6000}\sqrt{n} \le k_3\sqrt{n}$.

Step 2: $d_F(G^{0,m}, G^0) \le k_1$.

Since $\sigma \le C_3\sqrt{n/\log n}$, using theorem 3.3 we get

$$d_\infty(\widetilde{G}, G^\star) = \|\widetilde{G} - G^\star Q\|_\infty \le C_0C_3 \le \frac{1}{2}. \tag{65}$$

Using Weyl's singular-value inequality [5, Chapter III] we have

$$\begin{aligned}\sigma_r(\widetilde{G}_i) &\geq \sigma_r(G_i^\star) - \sigma_1(G_i^\star Q - \widetilde{G}_i)\\ &= 1 - \|\widetilde{G}_i - G_i^\star Q\|_2\\ &\geq 1 - d_\infty(\widetilde{G}, G^\star) \geq \frac{1}{2}.\end{aligned} \tag{66}$$

Therefore, $\min_{i\in[n]} \sigma_r(\widetilde{G}_i) \geq \frac{1}{2}$. Using corollary C.8, we derive

$$d_F(G^0, G^{0,m}) = d_F(\pi^n\widetilde{G}, \pi^n\widetilde{G}^{(m)}) \leq 8 d_F(\widetilde{G}, \widetilde{G}^{(m)}) \tag{67}$$

Using Weyl's inequality [5, Chapter III]:

$$\begin{aligned}|\lambda_r(C^{(m)}) - n| &\leq \sigma\|W^{(m)}\|_2 \leq \sigma C_2\sqrt{n},\\ |\lambda_{r+1}(C^{(m)}) - 0| &\leq \sigma\|W^{(m)}\|_2 \leq \sigma C_2\sqrt{n}.\end{aligned} \tag{68}$$

Using Davis–Kahan [8],

$$d_F(\widetilde{G}, \widetilde{G}^{(m)}) \leq \frac{\sqrt{2}\sigma\|\Delta W^{(m)}\widetilde{G}^{(m)}\|_F}{n - 3\sigma C_2\sqrt{n}}. \tag{69}$$

Since $\widetilde{G}^{(m)}$ and $\Delta W^{(m)}$ are independent, we choose $U^{(m)} = \{\widetilde{G}^{(m)}\}$ in lemma C.2, then

$$\forall m \in [n], \|\Delta W^{(m)}\widetilde{G}^{(m)}\|_F \leq C_1\sqrt{n\log n} + C_1\sqrt{n}\|\widetilde{G}_m^{(m)}\|_F \tag{70}$$

Notice that

$$C^{(m)}\widetilde{G}^{(m)} = G^\star G^{\star\top}\widetilde{G}^{(m)} + \sigma W^{(m)}\widetilde{G}^{(m)} = \widetilde{G}^{(m)}\Lambda, \tag{71}$$

where $\Lambda = diag(\lambda_1(C^m), \cdots, \lambda_r(C^m))$. $\forall m \in [n]$,

$$\begin{aligned}\|\widetilde{G}_m^{(m)}\| &= \|G_m^\star G^{\star\top}\widetilde{G}^{(m)}\Lambda^{-1}\|_F\\ &\leq \|G_m^\star\|_2\|G^{\star\top}\|_2\|\widetilde{G}^{(m)}\|_F\|\Lambda^{-1}\|_2\\ &\leq \sqrt{n}\sqrt{nr}\frac{1}{\lambda_r(C^{(m)})} \leq 2\sqrt{r}.\end{aligned} \tag{72}$$

Combining (69),(70) and (72), we have

$$d_F(\widetilde{G}, \widetilde{G}^{(m)}) \leq \frac{\sqrt{2}\sigma\cdot 3\sqrt{r}C_1\sqrt{n\log n}}{n - 3\cdot\frac{n}{12000\sqrt{2}}} \leq \frac{\sqrt{2}}{12000}. \tag{73}$$

Substituting (73) into (67), we derive

$$d_F(G^0, G^{0,m}) \leq 8\times\frac{\sqrt{2}}{12000} < k_1. \tag{74}$$

Step 3: $\|WG^0\|_\infty \leq k_2\sqrt{n\log n}$.

Notice that $G^{0,m}$ is independent from $\Delta W^{(m)}$, we choose $U^{(m)} = \{G^{0,m}\}$ in lemma C.2, then

$$\forall m \in [n], \|\Delta W^{(m)}G^{0,m}\|_F \leq C_1\sqrt{n\log n} + C_1\sqrt{nr} \leq 2C_1\sqrt{r}\sqrt{n\log n}. \tag{75}$$

$\forall m \in [n]$, let $d_F(G^0, G^{0,m}) = \|G^0 - G^{0,m}Q_m\|_F$, then

$$\begin{aligned}\|W_m^\top G^0\|_F &= \|W_m^\top(G^0 - G^{0,m}Q_m) + W_m^\top G^{0,m}Q_m\|_F\\ &\leq C_2\sqrt{n}d_F(G^0, G^{0,m}) + \|W_m^\top G^{0,m}\|_F\\ &\leq C_2k_1\sqrt{n} + \|\Delta W^{(m)}G^{0,m}\|_F\\ &\leq C_2k_1\sqrt{n} + 2C_1\sqrt{r}\sqrt{n\log n}\\ &\leq k_2\sqrt{n\log n}.\end{aligned} \tag{76}$$

Therefore, $\|WG^0\|_\infty = \max_{1\leq m\leq n}\|W_m^\top G^0\|_F \leq k_2\sqrt{n\log n}$. □

Theorem C.10. *Under the condition of theorem C.9, with probability at least* $1 - O(n^{-2})$ *,*$\forall t \in \{0, 1, \cdots 3n^2 - 1\}$*, the following inequalities hold:*

$$d_F(G^t, G^\star) \le k_3\sqrt{n}, \tag{77}$$

$$d_F(G^{t,m}, G^t) \le k_1 \quad \forall m \in [n], \tag{78}$$

$$\|WG^t\|_\infty \le k_2\sqrt{n\log n}. \tag{79}$$

Proof. Let $T = 3n^2 - 1$. Choosing $U^{(m)} = \{G^{0,m}, G^{1,m}, \cdots, G^{T,m}\}$, we derive that with probability at least $1 - O(n^{-2})$,

$$\forall m \in [n], \max_{0\le t\le T} \|\Delta W^{(m)} G^{t,m}\|_F \le C_1\sqrt{n\log n} + C_1\sqrt{r}\sqrt{n} \le 2C_1\sqrt{r}\sqrt{n\log n}. \tag{80}$$

By theorem C.9, the initial points satisfy the properties in theorem C.10. We then prove the theorem by induction. Assuming that the three inequalities hold for some $t \in \{0, 1, \cdots, T-1\}$, we prove the inequalities for $t + 1$.

Step 1: $d_F(G^{t+1}, G^\star) \le k_3\sqrt{n}$.

Let $L = \frac{1}{n}C$. Since $\min_{i\in[n]} \sigma_r(G_i^\star) = 1$, using corollary C.8 we get

$$d_F(G^{t+1}, G^\star) = d_F(\pi^n LG^t, \pi^n G^\star) \le 4d_F(LG^t, G^\star) \le 4\sqrt{n}d_\infty(LG^t, G^\star). \tag{81}$$

Then, we bound $d_\infty(LG^t, G^\star)$.

$$\begin{aligned} d_\infty(LG^t, G^\star) &= \min_{Q\in O(r)} \|LG^t - G^\star Q\|_\infty \\ &= \min_{Q\in O(r)} \|\frac{1}{n}G^\star(G^{\star\top}G^t - nQ) + \frac{1}{n}\sigma WG^t\|_\infty \\ &\le \frac{1}{n}\min_{Q\in O(r)} \|G^\star(G^{\star\top}G^t - nQ)\|_\infty + \frac{1}{n}\sigma\|WG^t\|_\infty \\ &= \frac{1}{n}\min_{Q\in O(r)} \|G^{\star\top}G^t - nQ\|_F + \frac{1}{n}\sigma\|WG^t\|_\infty. \end{aligned} \tag{82}$$

Since $\|WG^t\|_\infty$ is bounded by the induction assumption, we only needs to bound $\min_{Q\in O(r)} \|G^{\star\top}G^t - nQ\|_F$.

$$\begin{aligned} \min_{Q\in O(r)} \|G^{\star\top}G^t - nQ\|_F^2 &= \|G^{\star\top}G^t\|_F^2 + n^2 r - 2n\max_{Q\in O(r)} \langle Q, G^{\star\top}G^t\rangle \\ &= \|G^{\star\top}G^t\|_F^2 + n^2 r - 2n\|G^{\star\top}G^t\|_\star \\ &= \sum_{i=1}^r (n - \sigma_i(G^{\star\top}G^t))^2 \\ &\le (\sum_{i=1}^r n - \sigma_i(G^{\star\top}G^t))^2 \\ &= (nr - \|G^{\star\top}G^t\|_\star)^2, \end{aligned} \tag{83}$$

where the last inequality holds because $\sigma_i(G^{\star\top}G^t) \le \|G^{\star\top}G^t\|_2 \le n$. Notice that $d_F^2(G^\star, G^t) = \|G^\star\|_F^2 + \|G^t\|_F^2 - 2\|G^{\star\top}G^t\|_\star = 2nr - 2\|G^{\star\top}G^t\|_\star$. Combining the above expression with (82),(83) we derive that,

$$\begin{aligned} d_\infty(LG^t, G^\star) &\le \frac{1}{n}(nr - \|G^{\star\top}G^t\|_\star) + \frac{1}{n}\sigma\|WG^t\|_\infty \\ &\le \frac{1}{2n}d_F^2(G^\star, G^t) + \frac{1}{n}\sigma\|WG^t\|_\infty \\ &\le \frac{k_3^2}{2} + \frac{\sigma k_2\sqrt{n\log n}}{n} \\ &\le \frac{k_3^2}{2} + \frac{1}{24000}. \end{aligned} \tag{84}$$

Substituting (84) into (81), we have

$$d_F(G^{t+1}, G^\star) \le (2k_3^2 + \frac{1}{6000})\sqrt{n} \le k_3\sqrt{n}. \tag{85}$$

Step 2: $d_F(G^{t+1,m}, G^{t+1}) \le k_1 \quad \forall m \in [n]$.

Let $L^{(m)} = \frac{1}{n}(G^\star G^{\star\top} + \sigma W^{(m)})$.

$$\begin{aligned} d_F(G^{t+1,m}, G^{t+1}) &= d_F(\pi^n L^{(m)} G^{t,m}, \pi^n L G^t) \\ &\le d_F(\pi^n L^{(m)} G^{t,m}, \pi^n L G^{t,m}) + d_F(\pi^n L G^{t,m}, \pi^n L G^t). \end{aligned} \tag{86}$$

We use lemma C.3 to control the two terms in the right hand of the inequality respectively.

We first bound $d_F(\pi^n L^{(m)} G^{t,m}, \pi^n L G^{t,m})$. Taking $x = y = G^{t,m}$ and $L' = \frac{1}{n}(G^\star G^{\star\top} + \sigma W^{(m)})$ in lemma C.3, we verify the conditions in the lemma as follows.

$$d_F(G^{t,m}, G^\star) \le d_F(G^{t,m}, G^t) + d_F(G^t, G^\star) \le (k_1 + k_3)\sqrt{n}. \tag{87}$$

Let $d_F(G^{t,m}, G^t) = \|G^{t,m} - G^t Q\|_F$.

$$\begin{aligned} \|WG^{t,m}\|_\infty &= \|W(G^{t,m} - G^t Q) + WG^t Q\|_\infty \\ &\le \|WG^t\|_\infty + \max_{1\le m\le n} \|W_m^\top (G^{t,m} - G^t Q)\|_F \\ &\le \|WG^t\|_\infty + \max_{1\le m\le n} \|W_m\|_2 \cdot d_F(G^{t,m}, G^t) \\ &\le k_2\sqrt{n\log n} + C_2\sqrt{n}k_1 \\ &\le 2k_2\sqrt{n\log n}. \end{aligned} \tag{88}$$

$$\begin{aligned} \|W^{(m)} G^{t,m}\|_\infty &\le \|WG^{t,m}\|_\infty + \|\Delta W^{(m)} G^{t,m}\|_F \\ &\le k_2\sqrt{n\log n} + C_2 k_1\sqrt{n} + 2C_1\sqrt{r}\sqrt{n\log n} \\ &\le 2k_2\sqrt{n\log n}, \end{aligned} \tag{89}$$

where the second inequality comes from (80). Using lemma C.3, we have

$$\begin{aligned} \epsilon = \frac{(k_1+k_3)^2}{2} + \sigma \cdot 2k_2\sqrt{\log n/n} \le \frac{1}{2}. \\ d_F(\pi^n L^{(m)} G^{t,m}, \pi^n L G^{t,m}) &\le 8 d_F(L^{(m)} G^{t,m}, L G^{t,m}) \\ &\le 8\|(L^{(m)} - L)G^{t,m}\|_F \\ &= 8\|\Delta W^{(m)} G^{t,m}\|_F \cdot \frac{\sigma}{n} \\ &\le \frac{8\sigma}{n} 2C_1\sqrt{r}\sqrt{n\log n} \le \frac{k_1}{2}. \end{aligned} \tag{90}$$

We then bound $d_F(\pi^n L G^{t,m}, \pi^n L G^t)$. Taking $x = G^{t,m}, y = G^t$ and $L' = L$ in lemma C.3, we verify the conditions in the lemma as follows.

$$\begin{aligned} d_F(G^{t,m}, G^\star) \le (k_1 + k_3)\sqrt{n}, \quad d_F(G^t, G^\star) \le k_3\sqrt{n}, \\ \|WG^{t,m}\|_\infty \le 2k_2\sqrt{n\log n}, \quad \|WG^t\|_\infty \le k_2\sqrt{n\log n}. \end{aligned}$$

These inequalities are either proved in (a) or coming from the induction assumption. Using lemma C.3, $\epsilon \le \frac{1}{2}$, $\rho \le 8(2(k_1+k_3) + \frac{\sigma\|W\|_2}{n}) \le \frac{1}{2}$, we have

$$d_F(\pi^n L G^{t,m}, \pi^n L G^t) \le \frac{1}{2} d_F(G^{t,m}, G^t) \le \frac{k_1}{2}. \tag{91}$$

Combining (90), (91), we have:

$$d_F(G^{t+1,m}, G^{t+1}) \le \frac{k_1}{2} + \frac{k_1}{2} = k_1. \tag{92}$$

Step 3: $\|WG^{t+1}\|_\infty \le k_2\sqrt{n\log n}$.

$\|WG^{t+1}\|_\infty = \max_{m\in[n]}\|W_m^\top G^{t+1}\|_F$. Let $d_F(G^{t+1,m}, G^{t+1}) = \|G^{t+1,m} - G^{t+1}Q\|_F$. We derive that $\forall m \in [n]$,

$$\begin{aligned}\|W_m^\top G^{t+1}\|_F &\le \|W_m^\top(G^{t+1}Q - G^{t+1,m})\|_F + \|W_m^\top G^{t+1,m}\|_F \\ &\le \|W_m\|_2 d_F(G^{t+1,m}, G^{t+1}) + \|\Delta W^{(m)} G^{t+1,m}\|_F \\ &\le C_2\sqrt{n}k_1 + 2C_1\sqrt{r}\sqrt{n\log n} \\ &\le k_2\sqrt{n\log n}.\end{aligned} \tag{93}$$

Therefore, $\|W_m^\top G^{t+1}\|_F \le k_2\sqrt{n\log n}$. □

C.6 Proof of Theorem 3.6

Since the preceding high-probability argument controls only the first $3n^2$ iterates, we proceed in two parts. Theorem C.11 establishes contraction over this finite horizon. Theorem C.12 then uses a deterministic continuation argument to extend this contraction to all later iterations and establish convergence to a fixed point.

Theorem C.11. *Under the condition of theorem C.9, with probability at least* $1 - O(n^{-2})$, $d_F(G^{t+1}, G^t) \le \frac{1}{2} d_F(G^t, G^{t-1}), \forall t \in \{1, \cdots, 3n^2 - 1\}$, *therefore,* $d_F(G^{3n^2-1}, G^{3n^2-2}) \le (\frac{1}{2})^{3n^2-3} k_3\sqrt{n} \le \frac{k_3}{4}$

Proof. With probability at least $1 - O(n^{-2})$ the conclusions of theorem C.10 hold. Let $1 \le t \le 3n^2 - 1$.

$$\begin{aligned}&d_F(G^t, G^\star) \le k_3\sqrt{n}, \quad d_F(G^{t-1}, G^\star) \le k_3\sqrt{n}, \\ &\|WG^t\|_\infty \le k_2\sqrt{n\log n}, \quad \|WG^{t-1}\|_\infty \le k_2\sqrt{n\log n}.\end{aligned}$$

Using lemma C.3, $\epsilon = \frac{k_3^2}{2} + \sigma k_2\sqrt{\log n/n} \le \frac{1}{2}, \rho \le 8(2k_3 + \frac{\sigma\|W\|_2}{n}) \le \frac{1}{2}$.

$$d_F(G^{t+1}, G^t) = d_F(\pi^n LG^t, \pi^n LG^{t-1}) \le \frac{1}{2} d_F(G^t, G^{t-1}). \tag{94}$$

Using (94) repeatedly we have:

$$\begin{aligned}d_F(G^{3n^2-1}, G^{3n^2-2}) &\le \frac{1}{2^{3n^2-2}} d_F(G^1, G^0) \\ &\le \frac{1}{2^{3n^2-2}}(d_F(G^1, G^\star) + d_F(G^0, G^\star)) \\ &\le \frac{1}{2^{3n^2-2}} \cdot 2k_3\sqrt{n} \\ &= \frac{1}{2^{3n^2-3}} \cdot k_3\sqrt{n} \le \frac{k_3}{4}.\end{aligned} \tag{95}$$

□

Theorem C.12. *Under the condition of theorem C.9, with probability at least* $1 - O(n^{-2})$, *the following statements hold:*

$$\forall k \ge 0, d_F(G^{T+k}, G^{T+k-1}) \le 2^{-k} d_F(G^T, G^{T-1}), \tag{96}$$

where $T = 3n^2 - 1$. *In the metric space* $(O_{d,r}^n/\sim, d_F)$, $[G^t]$ *converges to* $[G^\infty]$. $G^\infty \in O_{d,r}^n$ *is a fixed point of* $\mathcal{P} = \pi^n C$ *and satisfies the following properties:*

$$d_F(G^\infty, G^\star) \le \frac{3}{2} k_3\sqrt{n}, \quad \|WG^\infty\|_\infty \le (k_2 + C_2 k_3)\sqrt{n\log n}, \tag{97}$$

$$\forall t \ge 0, d_F(G^{t+1}, G^\infty) \le \frac{d_F(G^t, G^\infty)}{2}, \tag{98}$$

$$\forall i \in [n], G_i^\infty \Lambda_i = (CG^\infty)_i, \quad \Lambda_i = n[(LG^\infty)_i^\top (LG^\infty)_i]^{\frac{1}{2}}. \tag{99}$$

Proof. **Step1:** We first prove (96) by induction. When $k = 0$, the statement is trivial. Now we assume that the property holds for all $0 \le k \le m$, $m \ge 0$. We then prove the statement for $k = m + 1$. We first verify the conditions in lemma C.3 for $x = G^{T+m}, y = G^{T+m-1}$ and $L = L'$.

$$\begin{aligned} d_F(G^{T+m}, G^\star) &\le d_F(G^{T+m}, G^{T-1}) + d_F(G^{T-1}, G^\star) \\ &\le \sum_{l=0}^{m} d_F(G^{T+l}, G^{T+l-1}) + k_3\sqrt{n} \\ &\le (\sum_{l=0}^{m} 2^{-l}) d_F(G^T, G^{T-1}) + k_3\sqrt{n} \\ &\le \frac{k_3}{2} + k_3\sqrt{n} \le \frac{3k_3}{2}\sqrt{n}, \end{aligned}$$

where the last inequality follows from $d_F(G^T, G^{T-1}) \le \frac{k_3}{4}$ in theorem C.11. Similarly, $d_F(G^{T+m-1}, G^\star) \le \frac{3k_3}{2}\sqrt{n}$.

Assume that $d_F(G^{T+m}, G^{T-1}) = \|G^{T+m} - G^{T-1}Q\|_F$, then

$$\begin{aligned} \|WG^{T+m}\|_\infty &= \|W(G^{T+m} - G^{T-1}Q) + WG^{T-1}Q\|_\infty \\ &\le \|W(G^{T+m} - G^{T-1}Q)\|_\infty + \|WG^{T-1}\|_\infty \\ &\le k_2\sqrt{n\log n} + \|W(G^{T+m} - G^{T-1}Q)\|_F \\ &\le k_2\sqrt{n\log n} + C_2\sqrt{n} d_F(G^{T+m}, G^{T-1}), \end{aligned} \tag{100}$$

where the second inequality follows from theorem C.11. By the induction assumption we have

$$d_F(G^{T+m}, G^{T-1}) \le \sum_{l=0}^{m} d_F(G^{T+l}, G^{T+l-1}) \le \frac{k_3}{4}\sum_{l=0}^{m} 2^{-l} \le \frac{k_3}{2}. \tag{101}$$

Combining (100) and (101), we have $\|WG^{T+m}\|_\infty \le (k_2 + C_2k_3)\sqrt{n\log n}$. Similarly, we have $\|WG^{T+m-1}\|_\infty \le (k_2 + C_2k_3)\sqrt{n\log n}$. Using lemma C.3, we derive

$$\begin{aligned} d_F(G^{T+m+1}, G^{T+m}) &= d_F(\mathcal{P}G^{T+m}, \mathcal{P}G^{T+m-1}) \\ &\le \frac{1}{2} d_F(G^{T+m}, G^{T+m-1}) \le \frac{1}{2^{m+1}} d_F(G^T, G^{T-1}), \end{aligned} \tag{102}$$

which completes the proof of (96). Therefore $[G^t]$ is a Cauchy sequence in $(O^n_{d,r}/\sim, d_F)$ and must converge to $[G^\infty] \in O^n_{d,r}/\sim$

Step 2: Now we prove (97). Since (96) holds for all $k \ge 0$, following the same proof in step 1, we can derive that $\forall m \ge 0$,

$$d_F(G^{T+m}, G^\star) \le \frac{3k_3}{2}\sqrt{n}, \quad \|WG^{T+m}\|_\infty \le (k_2 + C_2k_3)\sqrt{n\log n}. \tag{103}$$

By continuity we derive (97).

Step 3: We now prove (98). Combining (103),(97) and using lemma (C.3), we deduce that $\forall t \ge 0, d_F(\mathcal{P}G^t, \mathcal{P}G^\infty) \le \frac{1}{2} d_F(G^t, G^\infty)$.

$$\begin{aligned} d_F(\mathcal{P}G^\infty, G^\infty) &\le d_F(\mathcal{P}G^\infty, \mathcal{P}G^t) + d_F(G^{t+1}, G^\infty) \\ &\le \frac{1}{2} d_F(G^t, G^\infty) + d_F(G^{t+1}, G^\infty). \end{aligned}$$

Letting $t \to \infty$, we get $d_F(PG^\infty, G^\infty) = 0$. Therefore, $\forall t \ge 0$ we have

$$d_F(G^{t+1}, G^\infty) \le d_F(\mathcal{P}G^t, \mathcal{P}G^\infty) + d_F(\mathcal{P}G^\infty, G^\infty) \le \frac{1}{2} d_F(G^t, G^\infty). \tag{104}$$

Step 4: Lastly, we prove (99). Since $d_F(\mathcal{P}G^\infty, G^\infty) = 0$, there exist a matrix $Q \in O(r)$ such that $\pi^n LG^\infty = G^\infty Q$. $\forall i \in [n]$, $\pi(LG^\infty)_i = G^\infty_i Q$. Assume the SVD decomposition of $\pi(LG^\infty)_i$ is $\pi(LG^\infty)_i = UDV^\top$, where $U \in \mathbb{R}^{d_i \times r}, U^\top U = I_r, D \in \mathbb{R}^{r\times r}, V \in O(r)$.

$$\begin{aligned} (LG^\infty)_i &= UDV^\top = UV^\top VDV^\top \\ &= (\pi(LG^\infty)_i) \cdot [(LG^\infty)_i^\top (LG^\infty)_i]^{\frac{1}{2}} = G^\infty_i Q \cdot [(LG^\infty)_i^\top (LG^\infty)_i]^{\frac{1}{2}}. \end{aligned} \tag{105}$$

Multiplying both sides of (105) with $(G_i^\infty)^\top$ and summing from 1 to n, we get

$$\begin{aligned}(G^\infty)^\top LG^\infty &= \sum_{i=1}^n (G_i^\infty)^\top (LG^\infty)_i \\ &= \sum_{i=1}^n (G_i^\infty)^\top G_i^\infty Q \cdot [(LG^\infty)_i^\top (LG^\infty)_i]^{\frac{1}{2}} \\ &= Q \sum_{i=1}^n [(LG^\infty)_i^\top (LG^\infty)_i]^{\frac{1}{2}}.\end{aligned} \tag{106}$$

Then, we will prove that the matrix $(G^\infty)^\top LG^\infty$ is positive definite. Since $(G^\infty)^\top LG^\infty = \frac{1}{n}(G^\infty)^\top G^\star G^{\star\top} G^\infty + \frac{\sigma}{n}(G^\infty)^\top W G^\infty$. By the Rayleigh quotient bound [5, Chapter I], $\forall x \in \mathbb{R}^{r\times 1}$ with $\|x\|_2 = 1$,

$$x^\top (G^\infty)^\top LG^\infty x \geq \frac{\sigma}{n}\lambda_{min}((G^\infty)^\top W G^\infty) + \frac{1}{n}\lambda_{min}((G^\infty)^\top G^\star G^{\star\top} G^\infty). \tag{107}$$

We bound the two terms in the right hand of the the inequality respectively.

First, we bound $\lambda_{min}((G^\infty)^\top W G^\infty)$.

$$\lambda_{min}((G^\infty)^\top W G^\infty) \geq -\|(G^\infty)^\top W G^\infty\|_2 \geq -n\sqrt{r}\|W\|_2 \tag{108}$$

Second, we bound $\lambda_{min}((G^\infty)^\top G^\star G^{\star\top} G^\infty)$. Since $d_F(G^\infty, G^\star) \leq \frac{3}{2}k_3\sqrt{n}$, by proposition C.5, we have $(1 - \frac{(\frac{3}{2}k_3)^2}{2})n \leq \sigma_i(G^{\star\top} G^\infty) \leq n$. Therefore

$$\lambda_{min}((G^\infty)^\top G^\star G^{\star\top} G^\infty) = \sigma_r^2(G^{\star\top} G^\infty) \geq (1 - \frac{9}{8}k_3^2)^2 n^2. \tag{109}$$

Substituting (108) and (109) into (107), we have:

$$x^\top (G^\infty)^\top LG^\infty x \geq -\sigma\sqrt{r}\|W\|_2 + (1 - \frac{9}{8}k_3^2)^2 n \geq ((1 - \frac{9}{8}k_3^2)^2 - \frac{1}{12000\sqrt{2}})n > 0, \tag{110}$$

which implies that $(G^\infty)^\top LG^\infty$ is positive definite.

Let $X = (G^\infty)^\top LG^\infty$, $Y = \sum_{i=1}^n [(LG^\infty)_i^\top (LG^\infty)_i]^{\frac{1}{2}}$, then Y is positive semi-definite and $X = QY$. Since X is positive definite, we have $X = X^{\frac{1}{2}} X^{\frac{1}{2}}$. Therefore,

$$X^{-\frac{1}{2}} Q^\top X^{\frac{1}{2}} = X^{-\frac{1}{2}} Y X^{-\frac{1}{2}}. \tag{111}$$

$X^{-\frac{1}{2}} Y X^{-\frac{1}{2}}$ is positive semi-definite and $Q^\top$ is similar to $X^{-\frac{1}{2}} Y X^{-\frac{1}{2}}$, therefore all eigenvalues of Q are positive real numbers. Since $Q \in O(r)$, we have $Q = I_r$, which implies that G^∞ is a fixed point of $\mathcal{P}$. Substituting $Q = I_r$ into (105), we have

$$(CG^\infty)_i = G_i^\infty \cdot n[(LG^\infty)_i^\top (LG^\infty)_i]^{\frac{1}{2}}. \tag{112}$$

□

C.7 Proof of Theorem 3.8

Step 1: Verify optimality and uniqueness of G^∞ and tightness of SDP.

By lemma 3.7, in order to verify the optimality of G^∞, we only need to prove that $M = L(G^\infty, \Lambda)$ is positive semi-definite and of rank $nd - r$. Since $\forall i \in [n], G_i^\infty \Lambda_i = (CG^\infty)_i$, $\Lambda_i = n[(LG^\infty)_i^\top (LG^\infty)_i]^{\frac{1}{2}}$, it is easy to show that $MG^\infty = 0$. We can decompose space R^{nd} by $R^{nd} = span(G^\infty) \bigoplus span(G^\infty)^\perp$.

If we can show that $\forall v \in span(G^\infty)^\perp, \|v\|_2 = 1, v^\top Mv > 0$, then the proof is complete. The reasons are as follows. On the one hand, since $MG^\infty = 0$ and $\forall v \in span(G^\infty)^\perp, \|v\|_2 = 1, Mv \neq 0$, the solution space of linear equation $Mx = 0$ is exactly $span(G^\infty)$. Therefore, rank$(M) = nd - dim(span(G^\infty)) = nd - r$. On the other hand, $\forall x \in R^{nd}$, we have decomposition $x = u + v$,

where $u \in span(G^\infty), v \in span(G^\infty)^\perp$. Since $MG^\infty = 0$,$Mu = 0$. Therefore, $x^\top Mx = (u^\top + v^\top)M(u+v) = v^\top Mv \geq 0$, from which we know that M is positive semi-definite.

Now we show that $\forall v \in span(G^\infty)^\perp, \|v\|_2 = 1, v^\top Mv > 0$. We assume that $v^\top = \begin{pmatrix} v_1^\top & \cdots & v_n^\top \end{pmatrix}^\top$ and $v_i \in \mathbb{R}^{d_i \times 1}$.

$$\begin{aligned} v^\top Mv &= \{\sum_{i=1}^n v_i^\top [G_i^\infty \Lambda_i (G_i^\infty)^\top + \tau_i (I_{d_i} - G_i^\infty (G_i^\infty)^\top)] v_i\} - v^\top Cv \\ &= \sum_{i=1}^n \tau_i \|v_i\|_2^2 + \sum_{i=1}^n (v_i^\top G_i^\infty \Lambda_i (G_i^\infty)^\top v_i - \tau_i v_i^\top G_i^\infty (G_i^\infty)^\top v_i) - v^\top Cv \\ &\geq \sum_{i=1}^n \tau_i \|v_i\|_2^2 - v^\top Cv, \end{aligned} \tag{113}$$

where the inequality follows from the Rayleigh quotient bound for Λ_i, since $v_i^\top G_i^\infty \Lambda_i (G_i^\infty)^\top v_i = ((G_i^\infty)^\top v_i)^\top \Lambda_i (G_i^\infty)^\top v_i \geq \tau_i \|(G_i^\infty)^\top v_i\|_2^2 = \tau_i v_i^\top G_i^\infty (G_i^\infty)^\top v_i$.

First, we provide a upper bound for $v^\top Cv$. Let $d_F(G^\infty, G^\star) = \|G^\infty - G^\star Q\|_F$.

$$\begin{aligned} v^\top Cv &= v^\top G^\star G^{\star\top} v + \sigma v^\top W v \\ &\leq \|G^{\star\top} v\|_2^2 + \sigma \lambda_{max}(W) \\ &\leq \|G^{\star\top} v\|_2^2 + \sigma \|W\|_2, \end{aligned} \tag{114}$$

where the two inequalities follow from the Rayleigh quotient bound and $\lambda_{\max}(W) \leq \|W\|_2$. Since $v \in span(G^\infty)^\perp$,

$$\|G^{\star\top} v\|_2 = \|v^\top (G^\infty - G^\star Q)\|_2 \leq d_F(G^\infty, G^\star). \tag{115}$$

Combining (114) and (115) we have

$$v^\top Cv \leq d_F^2(G^\infty, G^\star) + \sigma \|W\|_2 \leq (\frac{3}{2} k_3 \sqrt{n})^2 + \sigma C_2 \sqrt{n} \leq (\frac{9}{4} k_3^2 + \frac{1}{12000\sqrt{2}}) n. \tag{116}$$

Second, we provide a lower bound for $\sum_{i=1}^n \tau_i \|v_i\|_2^2$. Since τ_i is the smallest eigenvalue of $\Lambda_i = n[(LG^\infty)_i^\top (LG^\infty)_i]^{\frac{1}{2}}$, $\tau_i = n\sigma_r((LG^\infty)_i)$. We only need to bound $\sigma_r((LG^\infty)_i)$. By Weyl's singular-value inequality [5, Chapter III], we derive

$$\sigma_r((LG^\infty)_i) \geq \frac{1}{n}\sigma_r(G_i^\star G^{\star\top} G^\infty) - \frac{\sigma}{n}\sigma_1((WG^\infty)_i) \geq \frac{1}{n}\sigma_r(G^{\star\top} G^\infty) - \frac{\sigma}{n}\|WG^\infty\|_\infty. \tag{117}$$

Since $d_F(G^\infty, G^\star) \leq \frac{3}{2} k_3 \sqrt{n}$, by proposition C.5, we have $\sigma_r(G^{\star\top} G^\infty) \geq (1 - \frac{9}{8}k_3^2)n$. Therefore,

$$\sigma_r((LG^\infty)_i) \geq (1 - \frac{9}{8}k_3^2) - \frac{\sigma}{n} \cdot 2k_2\sqrt{n \log n} \geq 1 - \frac{9}{8}k_3^2 - \frac{1}{12000}. \tag{118}$$

Combining (113),(116), (118), we get,

$$v^\top Mv \geq \left((1 - \frac{9}{8}k_3^2 - \frac{1}{12000}) - (\frac{9}{4}k_3^2 + \frac{1}{12000\sqrt{2}}) \right) n > 0, \tag{119}$$

which completes the proof of the first part of the theorem.

Step 2: $d_F(G^\infty, G^\star) \lesssim \sigma$.

Note that G^∞ is a global optimum we have

$$tr(G^{\star\top} C G^\star) = \sum_{i,j=1}^n tr(G_i^{\star\top} C_{ij} G_j^\star) \leq \sum_{i,j=1}^n tr(G_i^\infty C_{ij} G_j^\infty) = tr((G^\infty)^\top C G^\infty). \tag{120}$$

Using the model $C = G^\star G^{\star\top} + \sigma W$, we derive

$$n^2 r + \sigma tr(G^{\star\top} W G^\star) \leq tr\left((G^{\star\top} G^\infty)^\top \cdot G^{\star\top} G^\infty\right) + \sigma tr((G^\infty)^\top W G^\infty). \tag{121}$$

Assuming that the eigenvalues of $G^{\star\top}G^{\infty}$ are $\sigma_1, \cdots, \sigma_r$, we have

$$\sum_{i=1}^{r}(n^2-\sigma_i^2) \le \sigma\left(tr((G^{\infty})^{\top}WG^{\infty}) - tr(G^{\star\top}WG^{\star})\right). \tag{122}$$

Since $n^2-\sigma_i^2 = (n-\sigma_i)(n+\sigma_i) \ge n(n-\sigma_i)$, we have

$$\sum_{i=1}^{r}(n-\sigma_i) \le \frac{\sigma}{n}\left(tr((G^{\infty})^{\top}WG^{\infty}) - tr(G^{\star\top}WG^{\star})\right). \tag{123}$$

Assume that $d_F(G^{\star}, G^{\infty}) = \|G^{\infty} - G^{\star}Q\|_F$. The difference of quadratic terms can be bounded by

$$\begin{aligned} tr((G^{\infty})^{\top}WG^{\infty}) - tr(G^{\star\top}WG^{\star}) &= tr\left((G^{\infty}-G^{\star}Q)^{\top}W(G^{\infty}+G^{\star}Q)\right) \\ &\le \|(G^{\infty}-G^{\star}Q)^{\top}\|_F \cdot \|W(G^{\infty}+G^{\star}Q)\|_F \\ &\le d_F(G^{\star},G^{\infty})\cdot C_2\sqrt{n}\cdot 2\sqrt{nr} = 2C_2\sqrt{r}nd_F(G^{\star},G^{\infty}). \end{aligned} \tag{124}$$

Using the fact that $d_F^2(G^{\star},G^{\infty}) = 2(nr - \|G^{\star\top}G^{\infty}\|_{\star})$ and combining (123) and (124), we derive

$$\frac{1}{2}d_F^2(G^{\star},G^{\infty}) \le \frac{\sigma}{n}\cdot 2C_2\sqrt{r}nd_F(G^{\star},G^{\infty}). \tag{125}$$

Therefore, we get

$$d_F(G^{\star},G^{\infty}) \le 4C_2\sqrt{r}\sigma. \tag{126}$$

Step 3: $d_{\infty}(G^{\infty},G^{\star}) \lesssim \sigma\sqrt{\frac{\log n}{n}}$

Since G^{∞} is a fixed point of $\pi^n L = \pi^n \frac{C}{n}$, $G_i^{\infty} = \pi((LG^{\infty})_i)$. Let $d_F(G^{\infty},G^{\star}) = \|G^{\infty}-G^{\star}Q\|_F$. Using lemma C.7 we derive

$$\|G_i^{\infty} - G_i^{\star}Q\|_F = \|\pi((LG^{\infty})_i) - \pi(G_i^{\star}Q)\|_F \le 4\|(LG^{\infty})_i - G_i^{\star}Q\|_F. \tag{127}$$

Noticing that $(LG^{\infty})_i = \frac{1}{n}G_i^{\star}G^{\star\top}G^{\infty} + \frac{\sigma}{n}(WG^{\infty})_i$, we have

$$\begin{aligned} \|(LG^{\infty})_i - G_i^{\star}Q\|_F &= \|\frac{1}{n}G_i^{\star}(G^{\star\top}G^{\infty}-nQ) + \frac{\sigma}{n}(WG^{\infty})_i\|_F \\ &\le \frac{1}{n}\|G^{\star\top}G^{\infty}-nQ\|_F + \frac{\sigma}{n}\|WG^{\infty}\|_{\infty} \\ &\le \frac{1}{n}\|G^{\star\top}G^{\infty}-nQ\|_F + \frac{\sigma}{n}(k_2+C_2k_3)\sqrt{n\log n}. \end{aligned} \tag{128}$$

Now we bound $\|G^{\star\top}G^{\infty}-nQ\|_F$.

$$\begin{aligned} \|G^{\star\top}G^{\infty}-nQ\|_F^2 &= \|G^{\star\top}G^{\infty}\|_F^2 + n^2r - 2n\langle Q, G^{\star\top}G^{\infty}\rangle \\ &= \sum_{i=1}^{r}\sigma_i^2(G^{\star\top}G^{\infty}) + n^2r - 2n\|G^{\star\top}G^{\infty}\|_{\star} \\ &= \sum_{i=1}^{r}(n-\sigma_i(G^{\star\top}G^{\infty}))^2 \\ &\le (n-\sigma_r(G^{\star\top}G^{\infty}))\sum_{i=1}^{r}n-\sigma_i(G^{\star\top}G^{\infty}) \\ &\le \frac{9}{8}k_3^2n(nr-\|G^{\star\top}G^{\infty}\|_{\star}) \\ &= \frac{9}{16}k_3^2nd_F^2(G^{\star},G^{\infty}), \end{aligned} \tag{129}$$

where the second equality is because $Q = argmin_{Q\in O(r)}\|G^{\infty}-G^{\star}Q\|_F = argmax_{Q\in O(r)}\langle Q, G^{\star\top}G^{\infty}\rangle$. The second inequality comes from $d_F(G^{\star},G^{\infty}) \le \frac{3}{2}k_3\sqrt{n}$ and proposition C.5. Combining (128) and (129) we derive

$$\begin{aligned} \|(LG^{\infty})_i - G_i^{\star}Q\|_F &\le \frac{3}{4\sqrt{n}}k_3d_F(G^{\star},G^{\infty}) + \frac{\sigma}{n}(k_2+C_2k_3)\sqrt{n\log n} \\ &\le \frac{3}{4\sqrt{n}}k_3\cdot 4\sqrt{r}C_2\sigma + (k_2+C_2k_3)\sigma\sqrt{\log n/n}. \end{aligned} \tag{130}$$

Substitute the above equation into (127) we derive

$$\begin{aligned} d_\infty(G^\infty, G^\star) &\le \|G^\infty - G^\star Q\|_\infty \\ &\le 4(\frac{3}{4\sqrt{n}} k_3 \cdot 4\sqrt{r} C_2 \sigma + (k_2 + C_2 k_3)\sigma\sqrt{\log n/n}) \\ &\lesssim \sigma\sqrt{\log n/n}. \end{aligned} \tag{131}$$

D Proof of the random observation model

D.1 Effective-noise estimates for the random observation model

Recall that $\widetilde{W} = \Delta_\star/p + \sigma W_\star/p$ and set $C_p^{(m)} = G^\star G^{\star\top} + \widetilde{W}^{(m)}$, where $\widetilde{W}^{(m)}$ zeros the m-th block row and column of $\widetilde{W}$.

Lemma D.1. *If $p > \frac{c_0 \log n}{n}$, then with probability at least $1 - O(n^{-2})$, for all $m \in [n]$,*

$$\|W_\star\|_2 \le C_1\sqrt{np}, \quad \|W_\star^{(m)}\|_2 \le C_1\sqrt{np}, \quad \|\Delta W_\star^{(m)}\|_2 \le C_1\sqrt{np}, \quad \|(W_\star)_m\|_2 \le C_1\sqrt{np},$$
$$\|\Delta_\star\|_2 \le C_1\sqrt{np}, \quad \|\Delta_\star^{(m)}\|_2 \le C_1\sqrt{np}, \quad \|\Delta(\Delta_\star)^{(m)}\|_2 \le C_1\sqrt{np}, \quad \|(\Delta_\star)_m\|_2 \le C_1\sqrt{np},$$

where the notation follows that of Lemma C.1.

Proof of Lemma D.1. Write $E = (e_{ij})$, $N = \sum_i d_i$ and $D_E = \max_i \sum_j e_{ij} d_j$. For fixed block dimensions, Bernstein's inequality [28, Theorem 2.8.4] and $np \gtrsim \log n$ give $D_E \le Cnp$ with probability at least $1 - O(n^{-4})$. Conditional on the masks, the scalar entries of $W_\star$ on and above the diagonal are independent, centered, and sub-Gaussian. We use the moment comparison in [4, Proposition 2.1 and Lemma 2.2, together with the sub-Gaussian comparison in Corollaries 3.2–3.3]:

$$\big(\mathbb{E}[\|W_\star\|_2^{2k} \mid E]\big)^{1/(2k)} \le CN^{1/(2k)}\big(\sqrt{D_E} + \sqrt{k}\big), \qquad k \ge 2.$$

To justify its application to arbitrary centered noise, let $W_\star'$ be an independent copy conditional on E. Conditional Jensen's inequality first replaces $W_\star$ by $W_\star - W_\star'$. The latter has independent symmetric entries whose even moments, after a universal rescaling, are dominated by Gaussian moments. Expanding the even trace moment therefore bounds it by that of a Gaussian matrix with the same mask. The cited moment comparison then gives the displayed estimate. Taking $k = \lceil 4\log n \rceil$ and applying Markov's inequality proves $\|W_\star\|_2 \le C\sqrt{np}$ except with probability $O(n^{-4})$, including the failure of the degree event. This argument conditions on the shared masks and does not assume independence of the unconditioned masked entries.

For the sampling fluctuations, let $B = \mathrm{BlkDiag}(G_1^\star, \dots, G_n^\star)$ and let J_n be the $n \times n$ all-ones matrix. Then $B^\top B = I_{nr}$ and

$$\Delta_\star = B\big[(E - pJ_n) \otimes I_r\big] B^\top.$$

The centered bounded-entry norm bound in [4, Corollary 3.12 and Remark 3.13] gives $\|E - pJ_n\|_2 \le C(\sqrt{np} + \sqrt{\log n}) \le C\sqrt{np}$ with probability at least $1 - O(n^{-4})$, and hence the same bound holds for $\|\Delta_\star\|_2$. Finally, if P_m zeros the m-th block, then $A^{(m)} = P_m A P_m$, $\Delta A^{(m)} = A - P_m A P_m$, and $\|A_m\|_2 \le \|A\|_2$. Thus all remaining bounds follow deterministically, simultaneously for every m, after enlarging the constant. □

For each m, take either $U^{(m)} = \{\widetilde{G}^{(m)}\}$ or $U^{(m)} = \{G^{t,m} : 0 \le t \le 3n^2 - 1\}$, where the spectral matrix and GPM iterates are computed from $C_p^{(m)}$.

Lemma D.2 (Removed-block concentration)**.** *Assume that $\sigma \lesssim \sqrt{np/\log n}$ and $p \gtrsim \frac{\log n}{\sqrt{n}}$. Then, with probability at least $1 - O(n^{-2})$,*

$$\max_{u \in U^{(m)}} \|\Delta W_\star^{(m)} u\|_F \lesssim \sqrt{np \log n}, \quad \max_{u \in U^{(m)}} \|\Delta(\Delta_\star)^{(m)} u\|_F \lesssim \sqrt{np \log n}, \qquad \forall\, 1 \le m \le n.$$

We now prove Lemma D.2.

Proof roadmap. We first derive a conditional concentration estimate that applies to both components of the sampled noise. For the leave-one-out GPM iterates, the Stiefel constraint immediately supplies the required block bounds (Lemma D.3). For spectral initialization, a coarse concentration bound first controls the spectral blocks (Lemma D.4), which then yields the sharp concentration bound (Lemma D.5). Both cases use the same block decomposition and concentration inequalities below.

Write $\mathcal{E}_{\mathrm{op}}$ for the simultaneous operator-norm event in Lemma D.1. As in the rest of the proof, constants absorb the fixed dimensions. We use Bernstein's inequality for bounded variables and the sub-Gaussian Hoeffding inequality from [28, Theorems 2.8.4 and 2.6.2].

A common conditional estimate. Fix m, and let u be independent of the removed pairs $\{(e_{mk}, W_{mk}) : k \in [n]\}$, with $u^\top u = nI_r$. Set $L = \max_k \|u_k\|_F$ and $a = \|u_m\|_F$. For either $A = W_\star$ or $A = \Delta_\star$, the removed-block structure gives the exact identity

$$\|\Delta A^{(m)} u\|_F^2 = \left\| \sum_{k=1}^{n} A_{mk} u_k \right\|_F^2 + \sum_{k \neq m} \|A_{km} u_m\|_F^2. \tag{132}$$

We will show that, conditionally on u, for every $x \geq 1$, with probability at least $1 - Ce^{-x}$,

$$\left\| \sum_k (W_\star)_{mk} u_k \right\|_F \leq C\big(\sqrt{npx} + Lx\big), \qquad \left\| \sum_k (\Delta_\star)_{mk} u_k \right\|_F \leq C\big(\sqrt{npx} + Lx\big). \tag{133}$$

For the first bound, fix a column j of u and write $a_k = \|u_k(:, j)\|_2^2$. Then $\sum_k a_k = n$ and $a_k \leq L^2$. Bernstein's inequality [28, Theorem 2.8.4], applied to $(e_{mk} - p)a_k$, yields

$$\sum_k e_{mk} a_k \leq np + C\big(\sqrt{npL^2 x} + L^2 x\big) \leq C(np + L^2 x)$$

except on an event of probability at most $2e^{-x}$. Given u and these masks, each scalar entry of $\sum_k e_{mk} W_{mk} u_k$ is a sum of independent, mean-zero sub-Gaussian variables, with sum of squared sub-Gaussian norms at most $C \sum_k e_{mk} a_k$. The sub-Gaussian Hoeffding inequality [28, Theorem 2.6.2] therefore bounds this entry by $C(\sqrt{npx} + Lx)$. A union bound over the entries proves the first inequality in (133). This argument conditions on the shared mask within each block; it does not assume that the masked entries are independent. It also applies to the diagonal block, since the distinct entries in any fixed row of W_{mm} are independent.

For the second bound, set $S_{mk} = G_m^\star G_k^{\star\top}$. Since $\|S_{mk}\|_2 \leq 1$, the scalar coefficients $b_k = (S_{mk} u_k)_{ij}$ satisfy $|b_k| \leq L$ and $\sum_k b_k^2 \leq n$. Bernstein's inequality [28, Theorem 2.8.4] applied directly to $(e_{mk} - p)b_k$, followed by an entrywise union bound, proves the claim. For this sampling-fluctuation estimate, independence of u from the removed masks alone is sufficient.

For the second term of (132), no further concentration argument is needed on $\mathcal{E}_{\mathrm{op}}$:

$$\left(\sum_{k \neq m} \|A_{km} u_m\|_F^2 \right)^{1/2} \leq \|A_m\|_2 \|u_m\|_F \leq C\sqrt{np}\, a.$$

Consequently, on $\mathcal{E}_{\mathrm{op}}$ and the event in (133),

$$\|\Delta A^{(m)} u\|_F \leq C\big(\sqrt{npx} + Lx + \sqrt{np}\, a\big), \qquad A \in \{W_\star, \Delta_\star\}. \tag{134}$$

We will also use the coarser estimate

$$\left\| \sum_k (W_\star)_{mk} u_k \right\|_F \leq C\sqrt{nx}, \tag{135}$$

with conditional failure probability Ce^{-x}. It follows from the same Hoeffding argument using only $\sum_k e_{mk} a_k \leq n$, and requires no bound on L.

All these tail estimates are obtained by conditioning on u, not on $\mathcal{E}_{\mathrm{op}}$. Thus they remain valid when bounds on L or a hold only on an event determined by u. We take $x = c \log n$, with c large enough that each conditional failure probability is $O(n^{-6})$, and then take a union bound over the required matrices and indices. The failure probability of the single common event $\mathcal{E}_{\mathrm{op}}$ is added only once.

Leave-one-out iterates.

Lemma D.3 (Concentration for Stiefel iterates)**.** *For each m, let $U^{(m)} = \{u^{(m,\ell)} : 1 \le \ell \le N_m\}$, where $N_m \le 3n^2$ is deterministic. Suppose each $u^{(m,\ell)} \in O^n_{d,r}$ is independent of $\{(e_{mk}, W_{mk}) : k \in [n]\}$. If $p > (\log n)^2/n$, then, with probability at least $1 - O(n^{-2})$, simultaneously for all m,*

$$\max_{u \in U^{(m)}} \|\Delta W_\star^{(m)} u\|_F \le C_2\sqrt{np\log n},$$
$$\max_{u \in U^{(m)}} \|\Delta(\Delta_\star)^{(m)} u\|_F \le C_2\sqrt{np\log n}.$$

Proof. For every such u, $u^\top u = nI_r$ and $L = a = \sqrt{r}$. Apply (134) to both $A = W_\star$ and $A = \Delta_\star$, with $x = c\log n$. The terms Lx and $\sqrt{np}\,a$ are absorbed into $C\sqrt{np\log n}$. Union bounding the conditional failures over both noise components and at most $3n^3$ pairs (m,ℓ), and adding $\mathbb{P}(\mathcal{E}_{\rm op}^c) = O(n^{-2})$ once, proves both bounds with a common constant C_2. The sampling condition implies that of Lemma D.1 for sufficiently large n. □

Spectral initialization: coarse bound, block control, and refinement. Throughout the next two lemmas, assume $p \gtrsim \log n/\sqrt{n}$ and $\sigma \lesssim \sqrt{np/\log n}$. Let $\widetilde{G}$ and $\widetilde{G}^{(m)}$ contain the top r eigenvectors of C_p and $C_p^{(m)}$, respectively, normalized so that their Gram matrices are nI_r. Each $\widetilde{G}^{(m)}$ is independent of the removed masks and noise blocks.

Lemma D.4 (Coarse spectral concentration and block bounds)**.** *With probability at least $1-O(n^{-2})$, simultaneously for all m,*

$$\|\widetilde{G}_m^{(m)}\|_F \le \frac{8}{7}\sqrt{r}, \qquad \|\Delta W_\star^{(m)}\widetilde{G}^{(m)}\|_F \le C_3\sqrt{n\log n},$$
$$\max_i \|\widetilde{G}_i^{(m)}\|_F \le \frac{C_4}{\sqrt{p}} \lesssim \frac{n^{1/4}}{\sqrt{\log n}}.$$

Proof. We first obtain the bound on the m-th block directly from the eigenvector equation. On $\mathcal{E}_{\rm op}$, the assumed noise and sampling levels give

$$\max\{\|\widetilde{W}\|_2,\ \max_m \|\widetilde{W}^{(m)}\|_2,\ \max_m \|\Delta\widetilde{W}^{(m)}\|_2\} \le C(1+\sigma)\sqrt{n/p} \le n/8, \tag{136}$$

with the constants chosen as in the spectral-initialization argument. Weyl's inequality [5, Chapter III] gives $\lambda_r(C_p^{(m)}) \ge 7n/8$ and $\lambda_{r+1}(C_p^{(m)}) \le n/8$, and the same bounds hold for C_p. Let Λ_m contain the top r eigenvalues of $C_p^{(m)}$. Its eigenvector equation is

$$\begin{aligned}\widetilde{G}^{(m)}\Lambda_m &= C_p^{(m)}\widetilde{G}^{(m)} \\ &= G^\star G^{\star\top}\widetilde{G}^{(m)} + \frac{1}{p}\Delta_\star^{(m)}\widetilde{G}^{(m)} + \frac{\sigma}{p}W_\star^{(m)}\widetilde{G}^{(m)}.\end{aligned} \tag{137}$$

The two noise terms have zero m-th block, so

$$\begin{aligned}\widetilde{G}_m^{(m)} &= G_m^\star G^{\star\top}\widetilde{G}^{(m)}\Lambda_m^{-1}, \\ \|\widetilde{G}_m^{(m)}\|_F &\le \frac{\|G^{\star\top}\|_2\|\widetilde{G}^{(m)}\|_F}{7n/8} = \frac{8}{7}\sqrt{r}.\end{aligned} \tag{138}$$

The coarse estimate (135), with $x = c\log n$, bounds the central block of $\Delta W_\star^{(m)}\widetilde{G}^{(m)}$ by $C\sqrt{n\log n}$. The remaining blocks contribute at most $C\sqrt{np}\|\widetilde{G}_m^{(m)}\|_F$ on $\mathcal{E}_{\rm op}$. Therefore, (132) and a union bound over m give, on a common event of probability $1 - O(n^{-2})$,

$$\|\Delta W_\star^{(m)}\widetilde{G}^{(m)}\|_F \le C_3\sqrt{n\log n}, \qquad \forall m. \tag{139}$$

The failure probability of $\mathcal{E}_{\rm op}$ is counted only once. We work on this event for the rest of the proof.

The Davis–Kahan theorem [8], (136), and Lemma D.1 now give

$$\begin{aligned} d_F(\widetilde{G}^{(m)}, \widetilde{G}) &\le \frac{C}{n}\|\Delta\widetilde{W}^{(m)}\widetilde{G}^{(m)}\|_F \\ &\le \frac{C}{np}\left(\|\Delta(\Delta_\star)^{(m)}\|_2\sqrt{nr} + \sigma\|\Delta W_\star^{(m)}\widetilde{G}^{(m)}\|_F\right) \\ &\le C\left(\frac{1}{\sqrt{p}} + \frac{\sigma\sqrt{n\log n}}{np}\right) \le \frac{C}{\sqrt{p}}, \end{aligned} \tag{140}$$

where we used (139) and $\sigma \lesssim \sqrt{np/\log n}$.

Next, for $i \neq m$, compare $\widetilde{G}^{(m)}$ with $\widetilde{G}^{(i)}$, aligning their columns by an optimal common rotation. Writing $A_{i,:}$ for the i-th block row of A, we obtain

$$\begin{aligned} \|(W_\star^{(m)})_{i,:}\widetilde{G}^{(m)}\|_F &\le \|(W_\star)_{i,:}\widetilde{G}^{(i)}\|_F + \|(W_\star)_{im}\widetilde{G}_m^{(m)}\|_F \\ &\quad + \|(W_\star)_{i,:}\|_2\, d_F(\widetilde{G}^{(m)}, \widetilde{G}^{(i)}) \\ &\le \|\Delta W_\star^{(i)}\widetilde{G}^{(i)}\|_F + C\sqrt{np} \\ &\quad + C\sqrt{np}\big(d_F(\widetilde{G}^{(m)}, \widetilde{G}) + d_F(\widetilde{G}^{(i)}, \widetilde{G})\big) \\ &\le C\sqrt{n\log n}. \end{aligned} \tag{141}$$

Here the last block term is bounded by $C\sqrt{np}$ using (138) and $\|(W_\star)_{im}\|_2 \le \|(W_\star)_m\|_2$; no additional concentration event is needed.

Finally, taking the i-th block of (137) gives

$$\begin{aligned} \widetilde{G}_i^{(m)} = \Big(&G_i^\star G^{\star\top}\widetilde{G}^{(m)} + \frac{1}{p}(\Delta_\star^{(m)})_{i,:}\widetilde{G}^{(m)} \\ &+ \frac{\sigma}{p}(W_\star^{(m)})_{i,:}\widetilde{G}^{(m)}\Big)\Lambda_m^{-1}. \end{aligned}$$

Since $\|\Lambda_m^{-1}\|_2 \le 8/(7n)$, $\|G_i^\star\|_2 = 1$, $\|G^{\star\top}\|_2 = \sqrt{n}$, and $\|\widetilde{G}^{(m)}\|_F = \sqrt{nr}$, we obtain

$$\begin{aligned} \|\widetilde{G}_i^{(m)}\|_F &\le \frac{8}{7}\sqrt{r} + \frac{C}{np}\left(\|\Delta_\star^{(m)}\|_2\sqrt{nr} + \sigma\|(W_\star^{(m)})_{i,:}\widetilde{G}^{(m)}\|_F\right) \\ &\le C\left(1 + \frac{1}{\sqrt{p}} + \frac{\sigma\sqrt{n\log n}}{np}\right) \le \frac{C}{\sqrt{p}}. \end{aligned}$$

Here we used Lemma D.1 and (141) for $i \neq m$; the case $i = m$ already follows from (138). All three conclusions hold on the same event. The final displayed rate follows from $p \gtrsim \log n/\sqrt{n}$. □

Lemma D.5 (Sharp spectral concentration)**.** *With probability at least $1 - O(n^{-2})$, simultaneously for all m,*

$$\begin{aligned} \|\Delta W_\star^{(m)}\widetilde{G}^{(m)}\|_F &\le C_5\sqrt{np\log n}, \\ \|\Delta(\Delta_\star)^{(m)}\widetilde{G}^{(m)}\|_F &\le C_5\sqrt{np\log n}. \end{aligned}$$

Proof. Apply (134) to $u = \widetilde{G}^{(m)}$, for both $A = W_\star$ and $A = \Delta_\star$, with $x = c\log n$. Lemma D.4 supplies $L \le C/\sqrt{p}$ and $a \le 8\sqrt{r}/7$. Hence, for either noise component,

$$\|\Delta A^{(m)}\widetilde{G}^{(m)}\|_F \le C\left(\sqrt{np\log n} + \frac{\log n}{\sqrt{p}} + \sqrt{np}\right) \le C_5\sqrt{np\log n},$$

where $p \gtrsim \log n/\sqrt{n}$ absorbs the middle term. For each m, the block bounds are determined by u, so the conditional tail estimate applies on that set of u's. Union bound over both noise components and all m, and add the failure probability of the common event from Lemma D.4 once. □

Completion of the proof of Lemma D.2. The leave-one-out iterates $G^{t,m}$, $0 \le t \le 3n^2 - 1$, are Stiefel matrices measurable with respect to the retained blocks, and hence independent of the removed masks and noise. Lemma D.3 therefore gives both required bounds for the iterates, and Lemma D.5 gives the corresponding bounds for $\widetilde{G}^{(m)}$. The sampling condition of Lemma D.2 implies those used above for sufficiently large n. Intersecting these events proves the two bounds simultaneously for all m, with probability $1 - O(n^{-2})$. □

D.2 Proof of Theorem 3.9

Using Lemma D.5, we have $\forall m \in [n]$,

$$\|\Delta\widetilde{W}^{(m)}\widetilde{G}^{(m)}\|_F \leq \frac{1}{p}\|\Delta(\Delta_\star)^{(m)}\widetilde{G}^{(m)}\|_F + \frac{\sigma}{p}\|\Delta W_\star^{(m)}\widetilde{G}^{(m)}\|_F \lesssim n. \tag{142}$$

Using Lemma D.3, we have

$$\forall m \in [n], \forall\, 0 \leq t \leq 3n^2 - 1, \|\Delta\widetilde{W}^{(m)}G^{t,m}\|_F \lesssim n, \tag{143}$$

where $G^{t,m}$ is the leave-one-out sequence generated by GPM with C replaced by $C_p^{(m)}$. Using lemma D.1 we have

$$\|\widetilde{W}\|_2 \lesssim n, \quad \|\widetilde{W}^{(m)}\|_2 \lesssim n, \quad \|\Delta\widetilde{W}^{(m)}\|_2 \lesssim n, \quad \|\widetilde{W}_m\|_2 \lesssim n. \tag{144}$$

These bounds supply the noise estimates needed by the full-observation argument, with the p-dependent scales established above. Once they hold, the trajectory-stability, contraction, recovery, and SDP-certification steps are deterministic. Applying those steps to $C_p = G^\star G^{\star\top} + \widetilde{W}$ proves Theorem 3.9; equality of the two noise distributions is neither asserted nor needed.

E Proof of the minimax lower bound

We prove Theorem 3.2 directly for rectangular Stiefel factors, including the rotational component. The argument does not require a synchronization lower bound as a black box. Let

$$\mathcal{G} = \prod_{i=1}^{n} O_{d_i,r}, \qquad h_i = r(d_i - r), \quad H = \sum_{i=1}^{n} h_i, \quad h_{\max} = \max_i h_i, \quad q = \frac{r(r-1)}{2}.$$

For $G \in \mathcal{G}$, let P_G be the law of the observed mask and measurements in the following Gaussian specialization of the random observation model. Take the independent upper-triangular scalar entries of W to have distribution $N(0, 1/4)$, so that $\mathbb{E}\exp(W_{ab}^2) = \sqrt{2} \leq 2$ and hence $\|W_{ab}\|_{\psi_2} \leq 1$. Thus, for $i < j$, an observed block has mean $G_i G_j^\top$ and independent $N(0, \sigma^2/4)$ noise entries. For a diagonal block, the entries on and above its diagonal have independent $N(0, \sigma^2/4)$ noise, and the lower triangle is obtained by symmetry. All unordered block masks are independent Bernoulli(p) variables and are independent of the noise. This fixed Gaussian normalization gives a subclass of the unit sub-Gaussian-norm model in the paper.

We allow arbitrary measurable estimators $\widehat{G} \in \mathbb{R}^{nd\times r}$ and define

$$R(n,p,\sigma) = \inf_{\widehat{G}} \sup_{G\in\mathcal{G}} \mathbb{E}_G\left[\frac{1}{n} d_F^2(\widehat{G}, G)\right]. \tag{145}$$

The quotient distance is a metric on orbits also for ambient matrices, so the argument below does not require $\widehat{G}$ to be Stiefel-valued. Throughout this appendix $n \geq 2$, $p \in (0,1]$, and $\sigma > 0$; at $\sigma = 0$ the asserted lower bounds are zero and hold trivially.

E.1 KL divergence with the observation mask

Lemma E.1 (Gaussian KL identity under Bernoulli sampling)**.** *If only off-diagonal blocks are observed, then*

$$D_{\mathrm{KL}}(P_G\|P_{G'}) = \frac{2p}{\sigma^2}\sum_{i<j}\|G_iG_j^\top - G_i'G_j'^\top\|_F^2. \tag{146}$$

If diagonal blocks are also observed, as in the paper, the right-hand side has the additional term

$$\frac{2p}{\sigma^2}\sum_i\sum_{a\leq b}\left[(G_iG_i^\top - G_i'G_i'^\top)_{ab}\right]^2. \tag{147}$$

In particular, if G and G' differ only at node k, then

$$D_{\mathrm{KL}}(P_G\|P_{G'}) \leq \frac{2p(n+3)}{\sigma^2}\|G_k - G_k'\|_F^2. \tag{148}$$

Proof. Write E for the observation mask. Its law is identical under both parameters. The chain rule for KL divergence therefore gives

$$D_{\mathrm{KL}}(P_G \| P_{G'}) = \mathbb{E}_E\big[D_{\mathrm{KL}}\big(P_G(C \mid E) \| P_{G'}(C \mid E)\big)\big].$$

Conditional on E, the distinct observed scalar coordinates are independent Gaussians of variance $\sigma^2/4$. The KL divergence between two such Gaussians is the squared difference of their means divided by $\sigma^2/2$. Summing over observed coordinates and using $\mathbb{E}e_{ij} = p$ proves (146)–(147). A diagonal block must be counted through its independent upper-triangular coordinates, not twice through its symmetric copies.

If only node k changes, each of its $n-1$ off-diagonal terms equals $\|G_k - G'_k\|_F^2$, since $G_j^\top G_j = I_r$. Furthermore,

$$\|G_k G_k^\top - G'_k G_k'^\top\|_F \le \|(G_k - G'_k)G_k^\top\|_F + \|G'_k(G_k - G'_k)^\top\|_F \le 2\|G_k - G'_k\|_F.$$

The sum over upper-triangular coordinates is at most the squared Frobenius norm. This proves (148); omitting diagonals only reduces the divergence. □

E.2 Quotient separation and the testing argument

Lemma E.2 (Anchored quotient separation)**.** *Partition $[n]$ into anchor nodes A and perturbed nodes P, with $|A| \ge |P|$. Suppose $G_i = G'_i = U_i \in O_{d_i,r}$ for $i \in A$, while $G_i, G'_i \in O_{d_i,r}$ are arbitrary for $i \in P$. Then*

$$d_F^2(G, G') \ge \frac{1}{2}\sum_{i \in P} \|G_i - G'_i\|_F^2. \tag{149}$$

Proof. Fix any $Q \in O(r)$. An anchor contributes $\|U_i - U_i Q\|_F^2 = \|I_r - Q\|_F^2$. At a perturbed node, write

$$G_i - G'_i Q = (G_i - G'_i) + G'_i(I_r - Q).$$

The inequality $\|X + Y\|_F^2 \ge \frac{1}{2}\|X\|_F^2 - \|Y\|_F^2$ follows by expanding $\frac{1}{2}\|X + 2Y\|_F^2 \ge 0$. Since $G_i'^\top G'_i = I_r$, summation yields

$$\sum_i \|G_i - G'_i Q\|_F^2 \ge \frac{1}{2}\sum_{i \in P} \|G_i - G'_i\|_F^2 + (|A| - |P|)\|I_r - Q\|_F^2.$$

The last term is nonnegative for every Q. Minimizing over Q proves the claim. In particular, the separation survives minimization over the common rotation. □

We record the elementary form of Assouad's argument used here, including its reduction from estimation to testing. For distributions P, Q, write $d_{\mathrm{TV}}(P, Q) = \sup_B |P(B) - Q(B)|$.

Lemma E.3 (Hypercube testing bound)**.** *Suppose parameters G^θ, indexed by $\theta \in \{-1, 1\}^M$, obey*

$$d_F^2(G^\theta, G^{\theta'}) \ge a\delta^2 \,\mathrm{Ham}(\theta, \theta')$$

and $d_{\mathrm{TV}}(P_\theta, P_{\theta'}) \le \alpha < 1$ whenever the Hamming distance is one. Then

$$R(n, p, \sigma) \ge \frac{a(1-\alpha)M\delta^2}{8n}. \tag{150}$$

Proof. For any estimator, choose a nearest hypercube point $G^{\widehat{\theta}}$ in d_F, breaking ties in a fixed order. The triangle inequality and nearest-point property imply

$$d_F(G^{\widehat{\theta}}, G^\theta) \le 2 d_F(\widehat{G}, G^\theta).$$

Thus its normalized squared loss is at least $a\delta^2 \,\mathrm{Ham}(\widehat{\theta}, \theta)/(4n)$. Average over the uniform prior on the cube. For each coordinate, pair every vertex with its bit-flipped neighbor. The sum of the two error probabilities of any binary test is at least $1 - d_{\mathrm{TV}}$, so the averaged error probability for that coordinate is at least $(1-\alpha)/2$. Sum over the M coordinates. The worst-case risk is at least this average, proving (150). □

In both constructions below, adjacent vertices differ only at one node and satisfy $\|G_k^\theta - G_k^{\theta'}\|_F \le 4\delta$. Lemma E.1 and $n+3 \le (5/2)n$ give

$$D_{\mathrm{KL}}(P_\theta \| P_{\theta'}) \le 80np\delta^2/\sigma^2. \tag{151}$$

We choose a universal $0 < \kappa \le 1/640$ and always take $\delta^2 \le \kappa\sigma^2/(np)$. Consequently the KL divergence is at most $1/8$. Pinsker's inequality [27, Chapter 2], $d_{\mathrm{TV}}(P,Q) \le \sqrt{D_{\mathrm{KL}}(P\|Q)/2}$, then gives $\alpha = 1/4$ in Lemma E.3.

E.3 The rectangular column-space component

Suppose $H > 0$. Let P consist of the $\lfloor n/2 \rfloor$ nodes with largest h_i, and put $A = [n] \setminus P$. Then $|A| \ge |P|$ and

$$M := \sum_{i \in P} h_i \ge \frac{\lfloor n/2 \rfloor}{n} H \ge H/3. \tag{152}$$

Set $U_i = (I_r; 0) \in O_{d_i,r}$. For $B \in \mathbb{R}^{(d_i - r)\times r}$ with $\|B\|_2 \le 1/2$, define

$$\Phi_i(B) = \begin{pmatrix} (I_r - B^\top B)^{1/2} \\ B \end{pmatrix}. \tag{153}$$

It satisfies $\Phi_i(B)^\top \Phi_i(B) = I_r$ and

$$\|B - B'\|_F \le \|\Phi_i(B) - \Phi_i(B')\|_F \le 2\|B - B'\|_F. \tag{154}$$

For completeness, put $S = (I_r - B^\top B)^{1/2}$ and $T = (I_r - B'^\top B')^{1/2}$. Both have smallest eigenvalue at least $\sqrt{3}/2$. The Sylvester identity

$$S(S-T) + (S-T)T = B'^\top B' - B^\top B$$

and its Frobenius inner product with $S - T$ imply

$$\sqrt{3}\|S - T\|_F \le \|B'^\top B' - B^\top B\|_F \le (\|B\|_2 + \|B'\|_2)\|B - B'\|_F \le \|B - B'\|_F.$$

Combining the upper and lower blocks proves (154).

For each $i \in P$, fill the h_i entries of B_i^θ with independent signs times δ, where

$$\delta^2 = \kappa \min\left\{ \frac{\sigma^2}{np}, \frac{1}{h_{\max}} \right\}. \tag{155}$$

Thus $\|B_i^\theta\|_2 \le \|B_i^\theta\|_F \le \sqrt{\kappa} \le 1/2$. Set $G_i^\theta = \Phi_i(B_i^\theta)$ on P and $G_i^\theta = U_i$ on A. Adjacent vertices obey the bound preceding (151). Lemma E.2 and the lower bound in (154) give

$$d_F^2(G^\theta, G^{\theta'}) \ge \tfrac{1}{2} \sum_{i \in P} \|B_i^\theta - B_i^{\theta'}\|_F^2 = 2\delta^2 \,\mathrm{Ham}(\theta, \theta'). \tag{156}$$

Equivalently, retaining only the lower blocks in the anchor proof yields a nonnegative rotation term $(|A| - \rho^2|P|)\|I_r - Q\|_F^2$, where $\rho = \max_i \|B_i^{\theta'}\|_2 \le 1/2$. This is the explicit anchored separation required before applying Assouad's argument.

Equations (152)–(156) and Lemma E.3 show that, for a universal $c > 0$,

$$R(n,p,\sigma) \ge c\frac{H}{n} \min\left\{ \frac{\sigma^2}{np}, \frac{1}{h_{\max}} \right\}. \tag{157}$$

When $H = 0$, we interpret the entire right-hand side as zero, without dividing by $h_{\max} = 0$.

E.4 The within-column-space rotational component

Suppose $q > 0$, equivalently $r \ge 2$. For $1 \le a < b \le r$, let

$$F^{ab} = \frac{e_a e_b^\top - e_b e_a^\top}{\sqrt{2}}.$$

These q matrices form a Frobenius-orthonormal basis of skew-symmetric matrices. For a skew-symmetric matrix S, define the Cayley transform

$$Q(S) = (I_r + S)(I_r - S)^{-1} = 2(I_r - S)^{-1} - I_r.$$

The inverse exists because $\|(I_r - S)x\|_2^2 = \|x\|_2^2 + \|Sx\|_2^2$. Direct multiplication using $S^\top = -S$ gives $Q(S)^\top Q(S) = I_r$. For skew-symmetric S, T of operator norm at most $1/2$, the resolvent identity gives

$$Q(S) - Q(T) = 2(I_r - S)^{-1}(S - T)(I_r - T)^{-1}.$$

Each inverse has operator norm at most one and smallest singular value at least $(1 + 1/4)^{-1/2}$. It follows that

$$\|S - T\|_F \le \|Q(S) - Q(T)\|_F \le 2\|S - T\|_F. \tag{158}$$

Choose any $|P| = \lfloor n/2 \rfloor$ perturbed nodes, with the others as anchors, and set

$$S_i^\theta = \delta \sum_{a<b} \theta_{i,ab} F^{ab}, \qquad \delta^2 = \kappa \min\left\{\frac{\sigma^2}{np}, \frac{1}{q}\right\}.$$

Then $\|S_i^\theta\|_2 \le \sqrt{q}\delta \le \sqrt{\kappa} \le 1/2$. Let $G_i^\theta = U_i Q(S_i^\theta)$ for $i \in P$, and $G_i^\theta = U_i$ for anchors. This cube has $M = |P|q \ge nq/3$ bits. The orthonormality of the basis, (158), and Lemma E.2 give the same separation $d_F^2 \ge 2\delta^2$ Ham. Adjacent points again obey $\|G_k^\theta - G_k^{\theta'}\|_F \le 4\delta$, so (151) applies. In this submodel the diagonal means $G_i^\theta G_i^{\theta\top} = U_i U_i^\top$ do not change at all. Applying Lemma E.3 therefore yields

$$R(n, p, \sigma) \ge cq \min\left\{\frac{\sigma^2}{np}, \frac{1}{q}\right\}. \tag{159}$$

For $r = 1$ there are no continuous rotational coordinates and this component is defined to be zero.

E.5 Combining the components and interpreting the bound

Both constructions are subfamilies of the same experiment. Its minimax risk is at least each subfamily's lower bound, and $\max\{x, y\} \ge (x + y)/2$. Consequently, after adjusting a universal constant,

$$R(n, p, \sigma) \ge c\left[\frac{H}{n} \min\left\{\frac{\sigma^2}{np}, \frac{1}{h_{\max}}\right\} + q \min\left\{\frac{\sigma^2}{np}, \frac{1}{q}\right\}\right], \tag{160}$$

with the zero-component conventions specified above. If

$$\frac{\sigma^2}{np} \max\{h_{\max}, q\} \le c_0$$

for a sufficiently small universal $c_0 > 0$, both minima choose $\sigma^2/(np)$. Equation (160) then proves Theorem 3.2:

$$R(n, p, \sigma) \ge c\frac{\sigma^2}{np}\left[\frac{1}{n}\sum_i r(d_i - r) + \frac{r(r-1)}{2}\right].$$

For homogeneous factors this dimension term is $r(d-r)+r(r-1)/2$. For fixed homogeneous dimensions with a positive dimension term, (160) implies that consistent recovery requires $\sigma^2/(np) \to 0$. Thus $\sqrt{np}$ is the statistical lower-bound noise scale, and the sufficient scale $\sqrt{np/\log n}$ matches it up to $\sqrt{\log n}$. More generally, the same conclusion holds for fixed $h_{\max}$ and q when $H/n + q$ is bounded away from zero. The local condition involves $h_{\max}$, rather than just the average dimension, to keep every perturbed node in its chart.

If $r = d_i = 1$ for every i, the parameter space is discrete and both continuous-dimension terms vanish; this construction makes no nontrivial claim for that case. Finally, the result is a necessary bound for statistical recovery. It is not an impossibility theorem for SDP tightness or for convergence of GPM, nor does it by itself establish achievability at the lower-bound scale.

F Broader Impacts

This paper presents work whose goal is to advance the field of machine learning. There are no negative social impacts.

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